CLASS X BIOLOGY — EXAM REVISION NOTES

10 chapters · labelled diagrams & photos · model answers · every past-paper question solved

Built from the **BCSE 2025**, **2024** and **2023** Biology question papers (Final I). Every Section A item and every Section B part in all three papers is answered inside.

How to use: Orange tags show where a topic appeared: P25 Q2a = 2025 paper question 2(a); P24 = 2024 paper; P23 = 2023 paper. "Sec A" = Q1 (MCQ, blanks, true/false, matching). Green boxes are **solved questions** with the mark value in brackets and the key answer in **red**. Purple tag diagram = the paper gave a figure. **Biology is a diagram subject:** every chapter has labelled drawings you can copy in the exam — practise redrawing them. Photos and real diagrams are from Wikimedia Commons (credit under each image).

⚠ **Read this first — limits of these notes** I had the three question papers only (no marking schemes, no textbook), so model answers follow the standard Class X (BCSE) syllabus and the wording of the questions. Where a figure in the paper was too small to read with certainty, or the answer depends on your textbook, it is marked **(check)**. Bhutan-specific facts (policies, Kuensel excerpts) are given as typical points; add the exact examples from your own textbook. Marks: write **exactly as many points as marks** (a 2-mark question needs two clear points).

How marks are given (all three papers)

- **Section A (40 marks):** Q1 only — (a) 25 MCQs, (b) 5 fill-in-the-blanks, (c) 5 true/false, (d) 5 matching. Do all of it; no negative marking, but circle only **ONE** letter.
- **Section B (60 marks):** attempt any **SIX** of Q2–Q9 (10 marks each, split into parts of 0.5–3 marks). Choose the questions with figures you can read.
- **"Label / identify"**: write the exact biological name — one word is enough.
- **"Draw and label"**: large, pencil, ruled label lines, a title, and only the parts asked. **Graphs**: title, axes labelled with units, even scale, points marked ×, smooth curve or ruled line.
- **"Justify / explain / comment"**: *claim + reason* ("because…"). **"Do you agree?"**: start Yes/No, then give the reasons asked for.
- **Variables:** independent = what you change; dependent = what you measure; controlled = what you keep the same.

Paper pattern (2023, 2024, 2025)

Part	Content	Marks
Q1(a)	25 MCQs (many use a figure, graph or table)	25
Q1(b)	Fill in the blanks	5
Q1(c)	True / False	5
Q1(d)	Match Column A with Column B	5
Sec B Q2–Q9	Any 6 of 8 long questions (figure study, data, graphs, case studies, drawing)	60

Topic map: where each chapter appeared

Chapter	2023	2024	2025
1 Cells & membrane transport	Q1a(i,iii,iv), Q3a, Q6a, Q8a	Q1a(i,ii), b(i), c(i), Q2a, Q3a	Q1a(i,ii,xx), d(i,ii), Q3b, Q5d, Q4b, Q6a
2 Photosynthesis	Q1a(v,vi), b(ii), c(i), Q3b, Q4d, Q6e	Q1a(iii,iv), c(ii), Q2b, Q8c	Q1a(iii,xvi,xxiv), c(i), Q2e, Q5a, Q7a, Q8a
3 Plant transport & hormones	Q1a(xviii), b(i), c(ii,iv), Q7d(i), Q9a,c	Q1a(v,vi,ix), b(iii,iv), Q7b	Q1a(iv,vii), b(i), Q4c
4 Digestion & nutrition	Q1a(vii,viii), b(iv), Q2b, Q4a(ii), Q6d, Q8b	Q1a(vii), b(ii), Q4a,b,c,d, Q9b	Q1a(v,ix,xvii,xxiii), c(ii), Q2a, Q9a
5 Blood & circulation	Q1a(ix), b(v), Q3c, Q4a(iii)	Q1a(viii), d(i,ii), Q6a,b	Q1a(vi), b(ii), Q6c, Q8b, Q9b
6 Excretion (kidney)	Q1a(xii), c(v), Q5a,c	Q1a(x), d(iii–v), Q6c	Q1a(viii), b(iii), Q7b,c
7 Nervous system	Q1a(x), Q4b	Q9a	Q1a(xxi), c(iii), Q9c
8 Cell division & heredity	Q1a(xix–xxii), d(i,ii), Q4c, Q5e, Q7c, Q9b,e	Q1a(xv–xx), b(v), Q7a, Q8a	Q1a(xi,xii,xix,xxv), b(ii), c(v), d(v), Q2c, Q4d,e, Q5e, Q7e, Q8c, Q9d
9 Evolution & biotechnology	Q1a(xxiii–xxv), d(iii–v), Q2d, Q5d, Q6c, Q7b, Q8d	Q1a(xxi–xxv), c(iv,v), Q3b, Q6d, Q8b	Q1a(xiii–xv), b(iv,v), d(iii,iv), Q2b,d, Q3d,e, Q5c, Q7d, Q8f
10 Ecology, biodiversity & health	Q1a(ii,xiii–xvii), b(iii), Q2c, Q5b, Q6b, Q8c,e, Q9d	Q1a(xi–xiv), c(iii), Q5, Q6a	Q1a(x,xviii,xxii), c(iv), Q3a,c,f, Q6d,e, Q8d,e, Q9e

Repeated every year: osmosis / potato experiments • photosynthesis factors & graphs • phloem vs xylem, ring experiment • digestion & H. pylori • ABO blood groups • nephron & urine formation • computer–nervous system analogy • mitosis vs meiosis • Punnett squares (earlobe, colour blindness) • natural selection, isolation, drift, Hardy–Weinberg • invasive species (lantana, crofton weed, polka dot), biodiversity and kidney/HIV lifestyle questions.

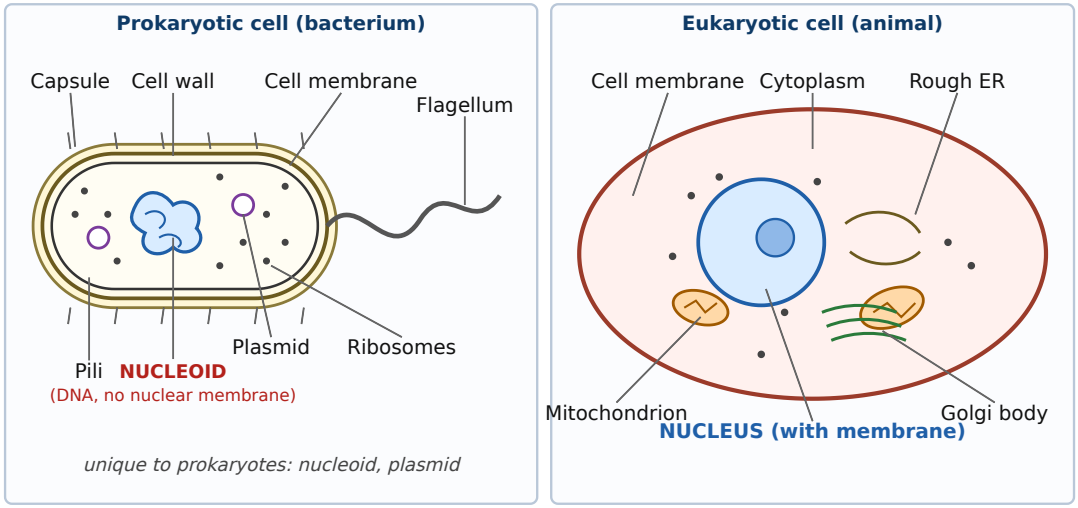
1 · CELLS & MEMBRANE TRANSPORT

P25 Q1a(i,ii,xx),d(i,ii),Q3b,Q4b,Q5d,Q6a · P24 Q1a(i,ii),b(i),c(i),Q2a,Q3a · P23 Q1a(i,iii,iv),Q3a,Q6a,Q8a

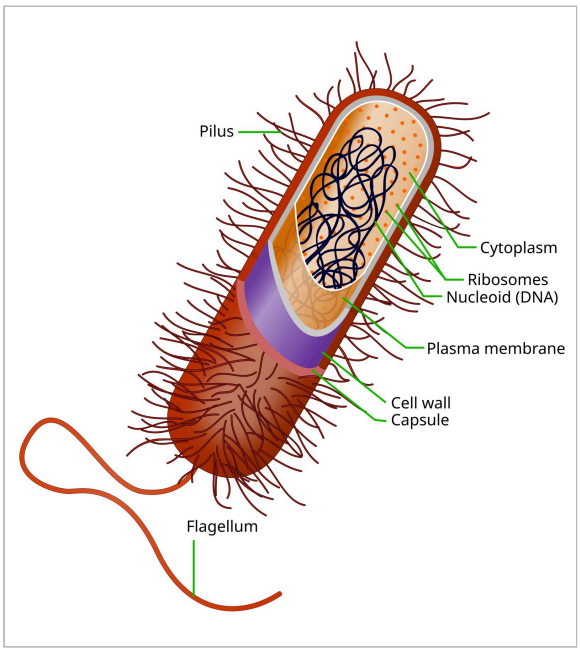
Prokaryotic vs eukaryotic cells

Feature	Prokaryote (bacteria)	Eukaryote (plant, animal)
Nucleus	**No true nucleus**: DNA lies free in a region called the **nucleoid**	**True nucleus** with a nuclear membrane
Organelles	**No membrane-bound organelles** (no mitochondria, ER, Golgi); only ribosomes	Many membrane-bound organelles
Size	Small, 1–10 µm (smaller than eukaryotic cells) P24 Q1c(i): FALSE	Larger, 10–100 µm
DNA	One circular chromosome + small rings (**plasmids**)	Several linear chromosomes
Cell wall	Peptidoglycan (bacteria)	Cellulose (plants); none in animals
Extras	Flagellum (movement), pili, capsule, mesosome	Mitochondria, chloroplast (plants)
Examples	E. coli, curd bacteria, H. pylori	Amoeba, yeast, plant and animal cells

Prokaryotic cell and eukaryotic cell



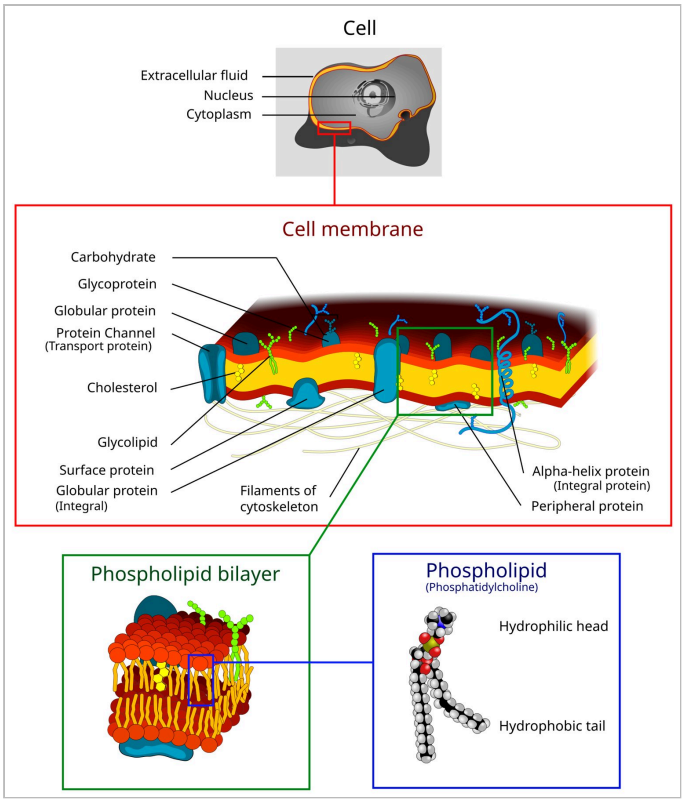
Prokaryotic cell (left, with the nucleoid unique to it) and an animal cell (right).



Structure of a typical bacterial (prokaryotic) cell.
Image: This vector image is completely made by..., CC BY-SA 4.0, Wikimedia Commons.

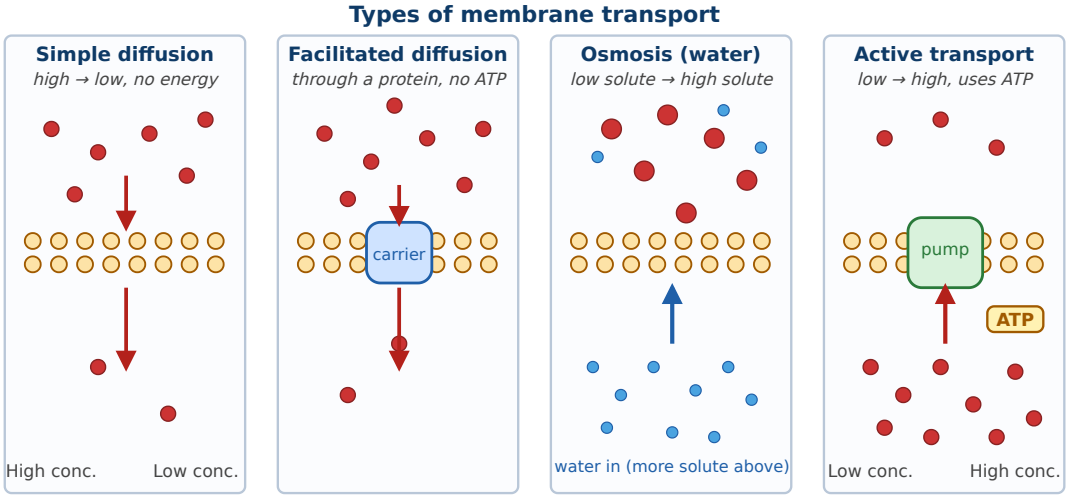
- Absent in a prokaryote: **mitochondria** P25 Q1a(i): C. Unique to prokaryotes: the **nucleoid** (and plasmid) P25 Q3b. A ciliated organism with a visible **nucleus** is eukaryotic P23 Q1a(i): B.
- **Why prokaryotes matter:** decomposers (recycle nutrients), nitrogen fixers, gut bacteria (digestion, vitamin K and B), curd and fermentation, **bioremediation**, biogas.

The cell membrane & types of transport



The cell membrane: a phospholipid bilayer with proteins (fluid mosaic model).
Image: derivative work: Dhatfield (talk) Cell_m..., CC BY-SA 3.0, Wikimedia Commons.

- The membrane is a **phospholipid bilayer with proteins** (fluid mosaic model); it is **selectively (semi-) permeable**: it lets some substances pass and blocks others.
- **Diffusion:** movement of molecules from **high to low concentration** (down the gradient); **no energy**. E.g. smell of *ema datsi* spreading in a room P24 Q1a(ii): D; O_2 and CO_2 exchange.
- **Facilitated diffusion:** diffusion of larger or charged molecules (glucose, ions) **through carrier / channel proteins**; down the gradient; **no ATP** P25 Q1d(ii): a.
- **Osmosis:** movement of **water** from a region of **low solute (high water) concentration to high solute concentration** through a semipermeable membrane P25 Q1a(ii): D.
- **Active transport:** movement **against the gradient** (low → high) using **carrier proteins and energy (ATP)**: e.g. mineral ions into root hairs, glucose uptake, Na^+–K^+ pump.



Passive (no energy): diffusion, facilitated diffusion, osmosis • Active: needs ATP and carrier protein, moves against the gradient

The four types of membrane transport side by side.

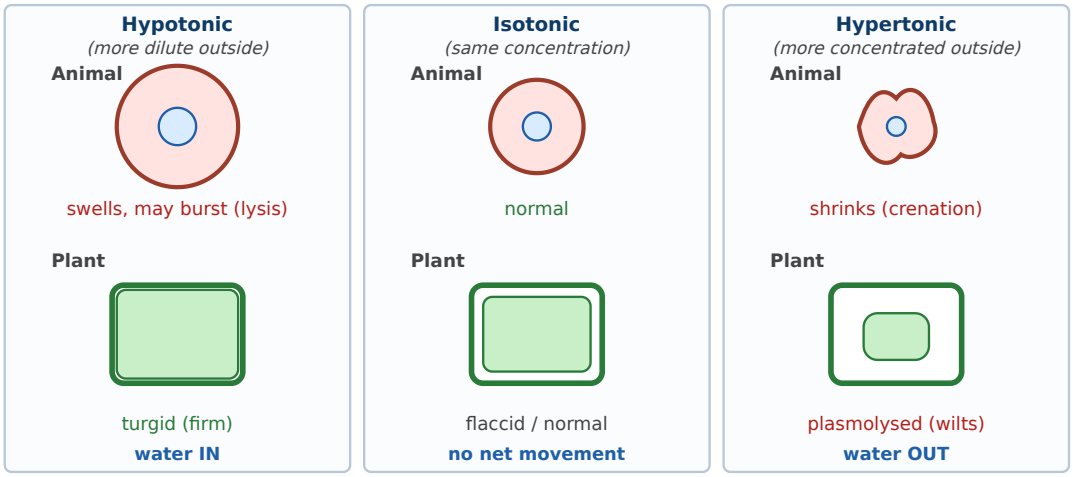
Tonicity: what happens to cells

Outside solution	Water movement	Animal cell	Plant cell

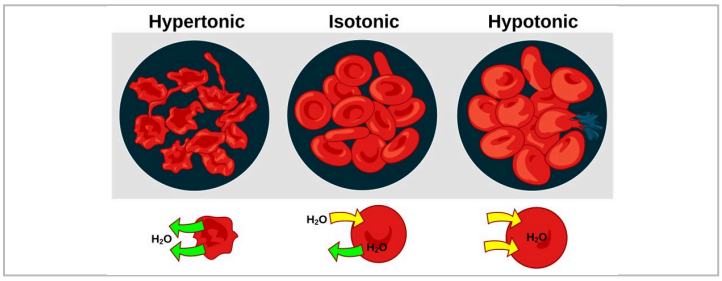
Hypotonic (more dilute than cell)	enters cell	swells, may burst	**turgid** (firm)

Isotonic (same)	no net movement	normal	normal / flaccid
Hypertonic (more concentrated)	leaves cell	shrinks	**plasmolysed** (wilts)

Effect of the outside solution on animal and plant cells



Effect of hypotonic, isotonic and hypertonic solutions on animal and plant cells.

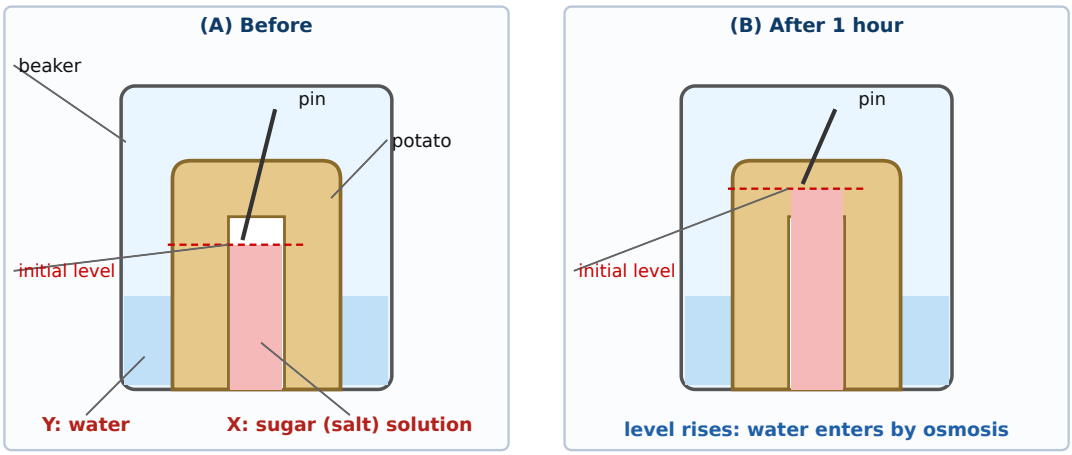


Real red blood cells in hypertonic, isotonic and hypotonic solutions.
Image: LadyofHats, Public domain, Wikimedia Commons.

- **Salting** fish, olives or vegetables: the **hypertonic** brine draws water out of bacteria, which die of dehydration **P24 Q1b(i): hypertonic**.
- **Wilting in salt water:** the soil water becomes hypertonic, so roots lose water by osmosis and leaves lose turgor **P25 Q6a**.
- **Soaking seeds:** water enters the seed by osmosis/imbibition; the seed coat softens and the embryo activates, so germination is faster **P23 Q6a**.
- **Wax on fruits** reduces water loss (transpiration) and membrane transport, so fruits stay fresh longer: statements I and III → **D** **P23 Q1a(iv)**.

Experiments you must know

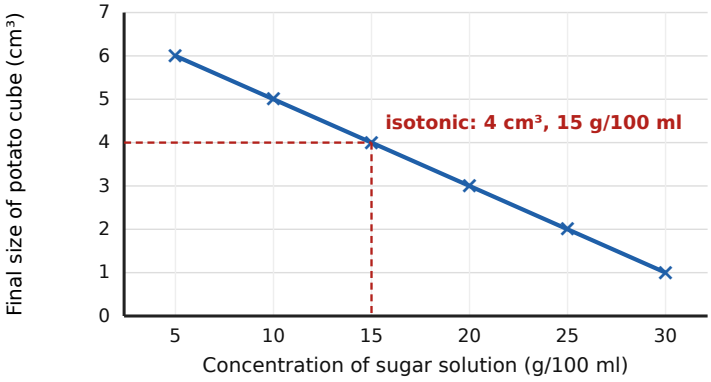
Potato osmoscope



Potato osmoscope: before (A) and after (B). Water enters the sugar solution, so its level rises.

- **Potato osmoscope:** a cavity is scooped in a potato and filled with **sugar (or salt) solution**; a **pin** marks the initial level; the potato stands in a beaker of **water**. After about 1 hour the level in the cavity **rises** because water moves by osmosis from the water (high water concentration) through the potato tissue (semipermeable) into the concentrated solution.
- **Control:** a **boiled (dead) potato**, or a cavity with water instead of sugar. Keep the same potato size, volume of liquid, temperature and time.
- **Potato cubes in sugar solutions:** cube **gains** volume in dilute solutions (hypotonic), **no change** in the isotonic one, and **shrinks** in strong solutions (hypertonic).

Potato cubes in sugar solutions (P25 Q4b)



Reading the graph
Above 4 cm³: cube gained water (hypotonic solution)
Equal to 4 cm³: isotonic (no net movement)
Below 4 cm³: cube lost water (hypertonic solution)

Final size of potato cubes against sugar concentration (2025 Q4b): the line crosses 4 cm^3 at 15 g/100 ml.

Solved questions

P25 Q3b [2] Prokaryotic cell: draw an **oval/rod** cell with **cell wall, cell membrane, cytoplasm, ribosomes, plasmid, flagellum** and a clearly labelled **nucleoid** (the DNA region with **no nuclear membrane**) — the label **nucleoid** is the part unique to prokaryotes (see figure).

P25 Q5d [2] The figure shows a **carrier protein with ATP** moving molecules from **low concentration (top) to high concentration (bottom)**. Type: **active transport**. Explanation: the molecules move **against the concentration gradient**, and the carrier protein uses **energy from ATP** to do so.

P25 Q4b [3] (i) **Solution C (15 g/100 ml)** is isotonic: the potato cube stays at **4 cm^3**, so there is **no net movement of water** — the sugar concentration outside equals that inside the cells. (ii) Graph: x-axis = concentration of sugar (g/100 ml) 5–30; y-axis = final size (cm^3) 0–7; plot (5,6) (10,5) (15,4) (20,3) (25,2) (30,1); join with a **ruled straight line sloping down**; title the graph.

P25 Q1d / Q6a Match: i Nucleoid (**c**); ii Carrier proteins = facilitated diffusion (**a**); iii waste-water treatment = bioremediation (**d**); iv foreign DNA = genetic modification (**b**); v recessive allele (**e**). Q6a: concentrated salt water is **hypertonic**, so root cells **lose water by osmosis**, cells lose turgor and the leaves **wilt**.

P24 Q2a [3] (i) a = **animal (eukaryotic) cell**; b = **bacterium (prokaryotic cell)**. (ii) Difference: cell a has a **true nucleus with a nuclear membrane**; cell b has **no true nucleus (nucleoid only)**. (Other: a has membrane-bound organelles, b does not.) (iii) Without prokaryotes: **dead organisms and waste would not decompose**, nutrients would not be recycled, nitrogen would not be fixed, so plants would not grow and the **ecosystem would collapse**.

P24 Q3a [5] Potato osmoscope. (i) **Aim:** to show that water moves into a concentrated solution through a semipermeable membrane by **osmosis** (using a potato). (ii) **X = sugar/salt solution** (in the cavity); **Y = water** in the beaker (check). (iii) Constant (control) variable: **same size/mass of potato**, volume of water, temperature or time. (iv) Salt makes the cavity solution **more concentrated (hypertonic)**, creating a gradient so water enters. (v) In a **hypotonic** solution the potato **absorbs water by osmosis, swells and becomes turgid** (gains mass).

P23 Q3a [3] (i) The potato cells act as a **semipermeable membrane**. The sugar solution is **more concentrated** than the water outside, so water moves in by **osmosis** and the level rises. (ii) **Control:** repeat with a **boiled potato** (membranes destroyed) or fill the cavity with **water only**, same size, volume, temperature and time — the level should **not rise**.

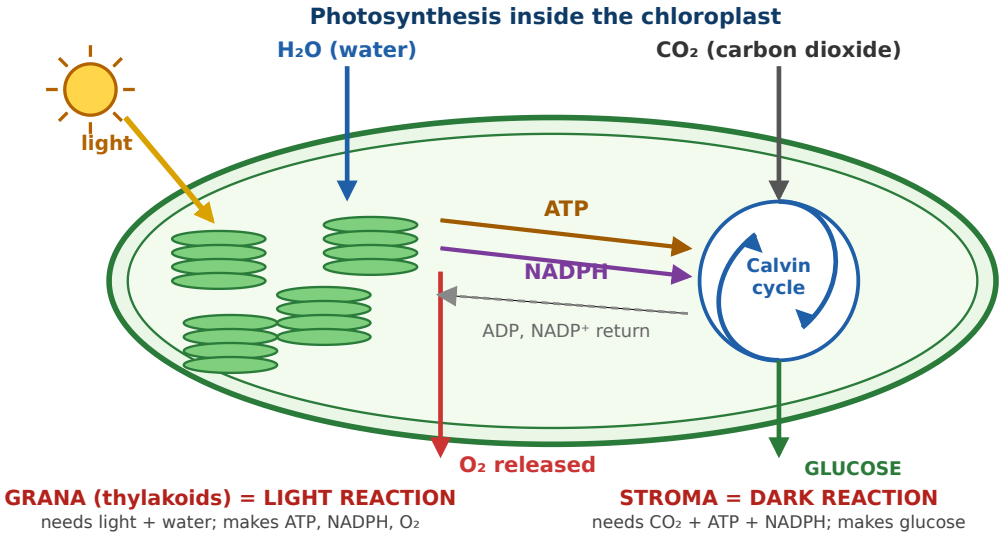
P23 Q8a [2] (i) **Process A: simple diffusion; Process B: facilitated diffusion** (check — the figure shows molecules moving down a gradient). (ii) In A molecules pass **directly through the lipid bilayer**; in B they pass **through carrier/channel proteins**. Neither uses ATP. (If B has ATP drawn, B = active transport.)

2 · PHOTOSYNTHESIS

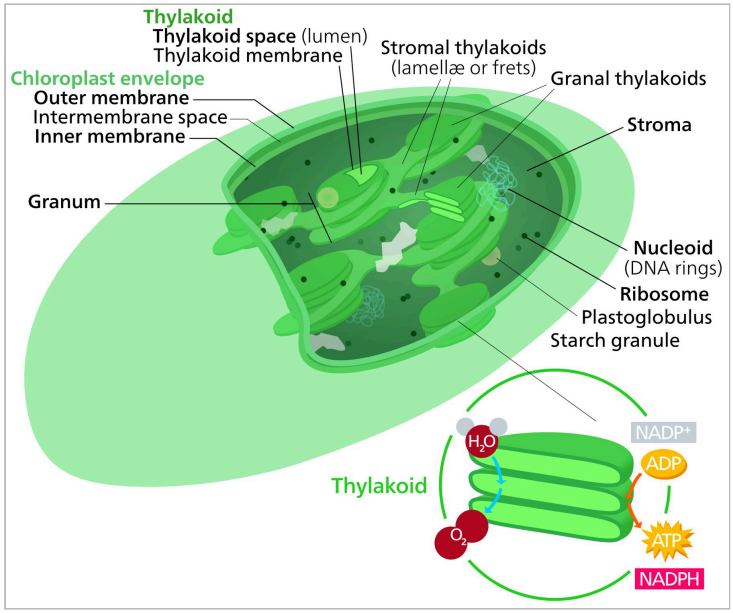
P25 Q1a(iii,xvi,xxiv),c(i),Q2e,Q5a,Q7a,Q8a · P24 Q1a(iii,iv),c(ii),Q2b,Q8c · P23 Q1a(v,vi),b(ii),c(i),Q3b,Q4d,Q6e

Key ideas

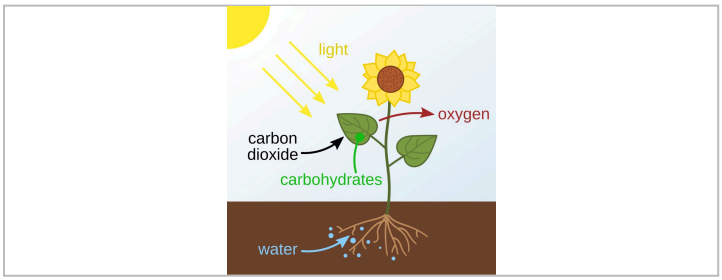
- **Equation:** $6CO_2 + 6H_2O \rightarrow$(light, chlorophyll)$\rightarrow C_6H_{12}O_6 + 6O_2$. Plants make glucose from CO_2 and water using light; photosynthesis **removes CO_2** and so helps to **minimise global warming** P23 Q1b(ii).
- **Chlorophyll** (in the chloroplast) **absorbs light energy** (mainly red and blue, reflects green) and changes it to chemical energy P25 Q2e.
- **Chloroplast:** double membrane; **grana** = stacks of **thylakoids** (hold chlorophyll) → **light reaction**; the fluid **stroma** → **dark reaction (Calvin cycle)** P24 Q2b(i).



Photosynthesis in the chloroplast: the light reaction in the grana makes ATP, NADPH and O_2; the Calvin cycle in the stroma makes glucose.



A chloroplast: grana (stacks of thylakoids) in the stroma.
Image: Chiswick Chap, CC BY-SA 4.0, Wikimedia Commons.



Photosynthesis in a plant: light, CO_2 and water in; carbohydrates and O_2 out (P24 Q1a(iii)).
Image: At09kg, Wattcle, Nefronus At09kg: origi..., CC BY-SA 4.0, Wikimedia Commons.

Light reaction vs dark reaction

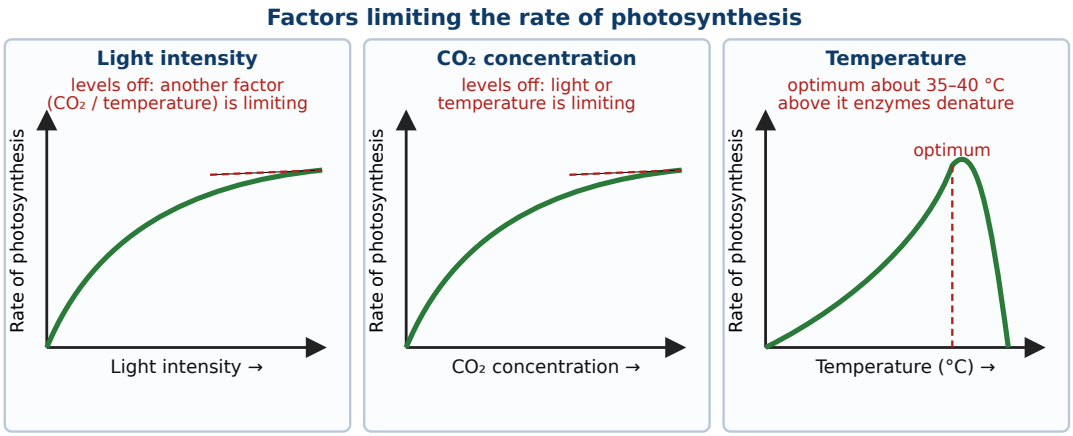
	Light reaction (1st phase)	Dark reaction / Calvin cycle (2nd phase)
Where	**Thylakoid membranes (grana)**	**Stroma**
Needs	**Light, water**, chlorophyll	**CO_2**, ATP, NADPH (enzymes)
What happens	Light excites chlorophyll; **water is split** (photolysis); **ATP** is formed; $NADP^+$ is changed to **NADPH**	CO_2 is **reduced to glucose**: **NADPH** gives hydrogen (reduces), **ATP** gives energy
Products	**O_2** (released), **ATP, NADPH**	**Glucose** (then starch, sucrose), ADP and $NADP^+$ are recycled

Light needed?	Yes	Light **independent**, but it **cannot continue without ATP and NADPH from the light reaction**

- Order of events in the light reaction: **II** splitting of water and **IV** formation of ATP → **C** P25 Q1a(xvi). Products passed to the Calvin cycle: **NADPH and ATP** → **D** P23 Q1a(v). Role of NADPH in the dark reaction: **reduces CO_2 to glucose** → **A** P25 Q1a(xxiv).
- Dark reaction can occur in day and night as it is light-independent → **TRUE** P24 Q1c(ii) **(check)**.

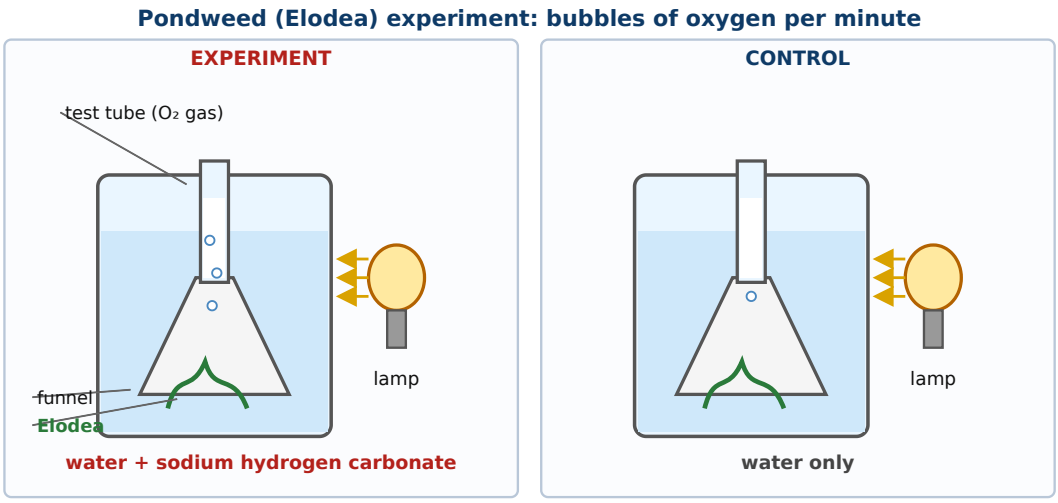
Factors affecting the rate of photosynthesis

- **External: light intensity**, **carbon dioxide** concentration, **temperature**, **water**. **Internal:** chlorophyll content, leaf structure. So **chlorophyll content is NOT external** → **B** P24 Q1a(iv) **(check)**.
- **Light intensity / CO_2:** rate rises then **levels off** (another factor becomes limiting). So "higher light always gives a higher rate regardless of other factors" is **FALSE** P25 Q1c(i).
- **Temperature:** rate rises to an **optimum** (about 35–40 °C), then **falls** because **enzymes denature**. So rate at 50 °C is **not** higher than at 20 °C → **FALSE** P23 Q1c(i). In cold weather, **temperature** is the limiting factor → **C** P23 Q1a(vi).
- **Limiting factor:** the factor in shortest supply, which controls the rate.



Typical graphs: light intensity, CO_2 concentration and temperature against rate of photosynthesis.

Experiments (Elodea / Hydrilla)



Pondweed experiment: count oxygen bubbles; sodium hydrogen carbonate supplies CO_2.

- Count **oxygen bubbles** (or measure the volume of O_2) given off by *Elodea* in a fixed time. **Light:** move the lamp nearer or farther. **Temperature:** water baths. **CO_2:** add sodium hydrogen carbonate ($NaHCO_3$).

- **Control** = the same setup **without the factor being tested**. In P25 Q1a(iii) the experiment has water + sodium hydrogen carbonate and the control has only water → the factor is **carbon dioxide (D)**.
- **Variables:** independent = factor you change; dependent = bubbles / volume of O_2 / rate; controlled = the other factors.

Solved questions

P25 Q2e [1] **Chlorophyll absorbs (traps) light energy** and converts it into chemical energy used in the light reaction to split water and make ATP and NADPH.

P25 Q5a [3] At high altitude it is cold, windy and the air is thin, so I would adjust: (1) **Temperature** — grow vegetables in a **greenhouse / plastic tunnel** to keep 25–35 °C, near the optimum for enzymes; (2) **Light** — use transparent covering, correct spacing so leaves do not shade each other, and add artificial light on cloudy days; (3) **Carbon dioxide and water** — add compost/manure (releases CO_2) and ventilate gently, irrigate regularly. Higher rate of photosynthesis → more food made → **higher yield**.

P25 Q7a [2] The chloroplast figure: **A = ATP** (made in light reaction and used in the Calvin cycle); **B = oxygen (O_2)** released when water is split; **C = carbon dioxide (CO_2)** entering the Calvin cycle; **D = grana / thylakoid** (site of light reaction) (check).

P25 Q8a [2] (i) Independent variable: **temperature**; dependent variable: **rate of photosynthesis**. (ii) The rate is maximum at **40 °C** (the peak of the curve), then falls as enzymes denature.

P24 Q2b [7] (i) Light reaction: **thylakoid membranes (grana)**; dark reaction: **stroma**. (ii) Light energy is **absorbed by chlorophyll**, which excites electrons; this energy **splits water** and is used to **make ATP and NADPH**. (iii) **ATP** supplies energy and **NADPH** supplies hydrogen to **reduce CO_2 to glucose** in the Calvin cycle; **O_2** is released as a by-product. (iv) Less water → **less water is split**, so **less ATP/NADPH/O_2** is made and the light reaction (and then the dark reaction) **slows**; stomata also close, reducing CO_2. (v) Design: **(1)** a **greenhouse** that keeps temperature near 30–35 °C and gives plenty of light (LED grow lights); **(2) CO_2 enrichment / good ventilation** and regular **irrigation** so no factor becomes limiting; use healthy, chlorophyll-rich varieties.

P24 Q8c [2] Graph: **independent variable = distance of the lamp from the pondweed (light intensity)**; **dependent variable = volume of oxygen produced in 24 hours (rate of photosynthesis)**. The curve falls as the lamp is moved away (less light, less photosynthesis).

P24 Q1a(iii) / P24 Q1c(ii) The sunflower figure (light, CO_2, water → carbohydrates and O_2): the phenomenon is photosynthesis, which can improve **crop yield (A)**. Dark reaction in day and night = **TRUE** (check).

P23 Q3b [3] Data: 25°C → 5, 30 → 7, 35 → 9, 40 → 7, 45 → 5 bubbles. (i) **Hypothesis:** as temperature rises to an optimum (35 °C) the rate of photosynthesis increases; above it the rate decreases. (ii) **Independent:** temperature; **dependent:** number of oxygen bubbles (per minute). (iii) **Graph:** x-axis temperature (°C) 25–45; y-axis number of bubbles 0–10; plot the five points and draw a smooth **hill-shaped curve peaking at 35 °C**; add title and units.

P23 Q4d [2] & Q6e [2] Greenhouse: (1) keep temperature at the **optimum** with heating/cooling and vents; (2) give **enough light** (transparent roof, artificial lights); (3) **raise CO_2** and supply **water** — so no factor limits and yield rises (any two). **Q6e:** the dark reaction does not need light itself, but it **uses the ATP and NADPH made by the light reaction**; without light these run out, so the dark reaction stops — the two phases depend on each other.

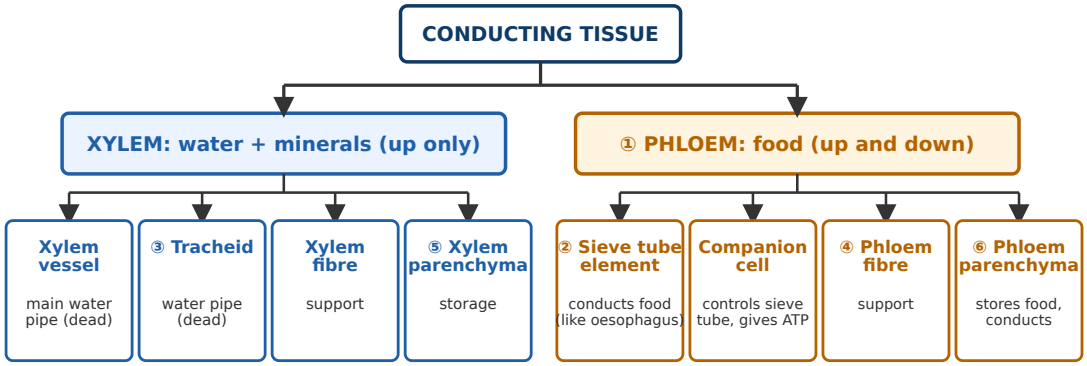
3 · TRANSPORT IN PLANTS & PLANT HORMONES

P25 Q1a(iv,vii),b(i),Q4c · P24 Q1a(v,vi,ix),b(iii,iv),Q7b · P23 Q1a(xviii),b(i),c(ii,iv),Q7d(i),Q9a,c

Conducting (vascular) tissues

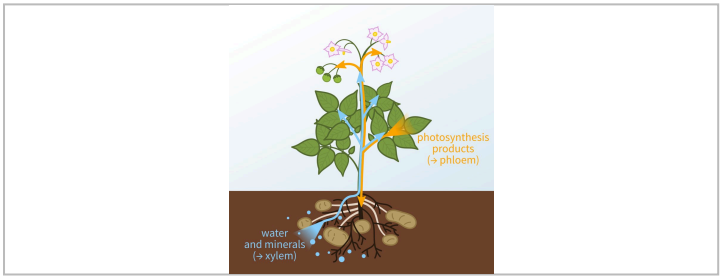
	Xylem	Phloem
Carries	**Water and minerals**	**Food** (sucrose, amino acids)
Direction	Roots → leaves (**upward only**)	Leaves → all parts (**up and down**)
Cells	**Dead**, hollow, lignified	Mostly **living** cells
Parts	**Vessels, tracheids, xylem fibres, xylem parenchyma**	**Sieve elements (tubes), companion cells, phloem fibres, phloem parenchyma**
Energy	Passive (transpiration pull)	Needs **energy (ATP)**

Conducting (vascular) tissue



Numbers ①–⑥ are the blanks of P24 Q7b: 1 phloem, 2 sieve element, 3 tracheid, 4 phloem fibre, 5 xylem parenchyma, 6 phloem parenchyma

Conducting tissue flow chart (completes P24 Q7b) and what each cell type does.



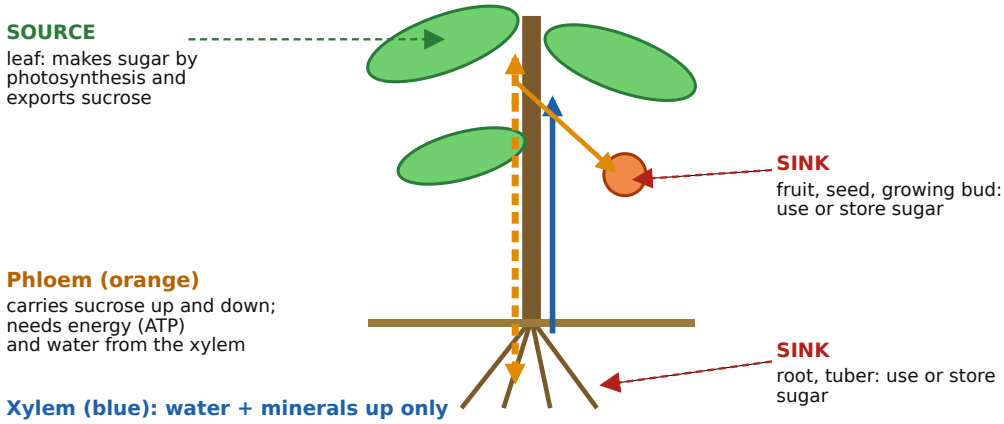
Xylem carries water and minerals up; phloem carries sugars to where they are used.
Image: Nefronus, CC BY-SA 4.0, Wikimedia Commons.

- **Sieve element:** long living tubes with sieve plates that **conduct food** — compared with the **oesophagus** that passes food along P24 Q1a(v): B. **Companion cell:** controls the sieve element and supplies ATP. **Phloem fibre:** support. **Phloem parenchyma: stores reserve food and conducts food** P25 Q1a(iv): D.

Translocation: source and sink

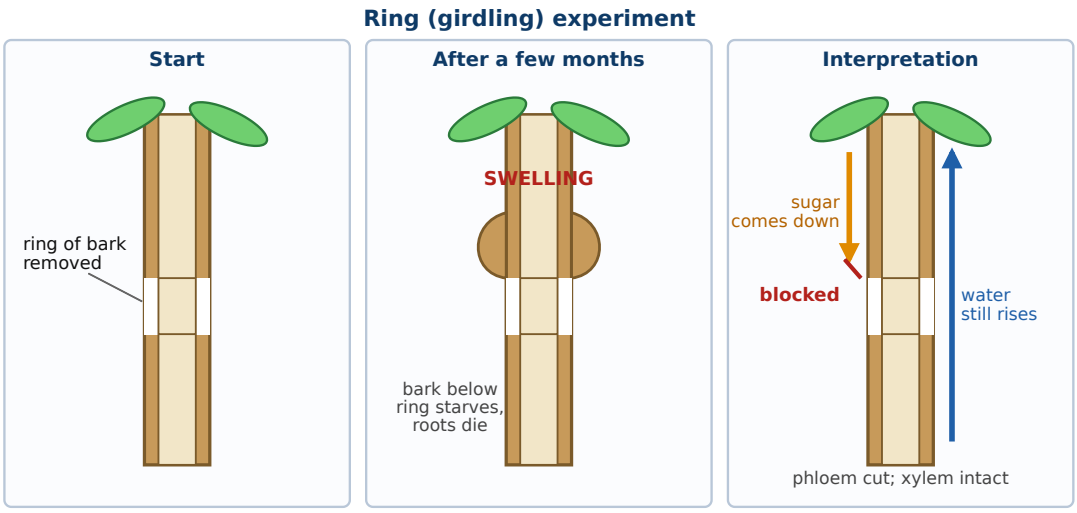
- **Translocation** = movement of the products of photosynthesis (sucrose) through the **phloem** P25 Q1b(i) P23 Q9c(i). It is driven by **active loading of sugars** into the phloem (needs ATP), after which water follows from the xylem by osmosis and pushes the sap to where it is used P24 Q1a(vi): C. So it **needs water and energy → TRUE** P23 Q1c(ii).
- **Source** = where food is made or released: **mature leaves** (also storage roots/tubers in spring). **Sink** = where food is used or stored: **roots, growing shoot tips, fruits, seeds, tubers** P23 Q9c(ii). Leaves are a source because they **make glucose by photosynthesis** in excess of their own needs P25 Q4c(ii).

Translocation: from source to sink through the phloem



Translocation: sucrose moves from the source (leaf) to the sink (root, fruit); water comes in from the xylem.

Ring (girdling) experiment



Ring of bark removed: food collects and the stem swells above the ring; below the ring and the roots starve.

- A complete ring of **bark (including the phloem)** is removed; the **xylem stays**. The part removed along with bark = **phloem** P24 Q1b(iv).
- After a few months: **swelling above the ring**, because sugars coming down from the leaves **cannot cross the gap** and pile up P23 Q7d(i). **Below the ring** the bark and the roots **do not get food**, so they die; the leaves stay green for a time because **water still rises in the xylem**.
- **Conclusion:** food is carried **downward through the phloem**, in the bark.



A tree with a ring of bark removed (girdled): the phloem is cut, so food cannot move down.
Image: Lamiot, CC BY 3.0, Wikimedia Commons.

Water loss & wilting

- On a hot sunny day leaf cells lose water by **transpiration** and become **flaccid**, so leaves droop, which reduces water loss P23 Q1b(i). The hormone **abscisic acid** closes stomata to reduce transpiration.

Plant hormones used in farming

Hormone	Functions / use
Auxin (IAA; synthetic **NAA**, **2,4-D**)	cell elongation; root growth in cuttings; **NAA** stops fruit drop and **prevents early sprouting of potatoes** P23 Q1c(iv): TRUE; **2,4-D = weedkiller** (kills broad-leaved weeds) P23 Q9a: X
Gibberellin	stem elongation, **breaks seed dormancy / promotes germination**, bigger fruits P23 Q9a: Z
Cytokinin	cell division, delays ageing of leaves
Ethylene	**early ripening of fruits and vegetables** P23 Q9a: Y = ethylene
Abscisic acid (ABA)	**closes stomata**, reduces transpiration in drought P25 Q1a(vii): C; seed and bud dormancy

- **Drawback of too much synthetic hormone:** unintended side effects and **negative environmental impact** (pollution of soil and water, harm to other plants) → **D** P24 Q1a(ix). The hormone commonly used to promote growth in agriculture: **auxin** (check textbook) P24 Q1b(iii).

Solved questions

P25 Q4c [3] (i) **Diagram:** draw a stem with the ring of bark removed; show the **stem swollen/bulging just above the ring**, **no change below**, leaves green, label “ring (gap)”, “swelling”, “xylem”, “phloem removed”. **Interpretation:** the phloem was cut, so sugar from the leaves **accumulates above the ring** and cannot reach the part below; this shows that **food is transported down by the phloem**, while the xylem (still intact) keeps carrying water up. (ii) Leaves are a **source** because they **manufacture glucose by photosynthesis** and export it.

P24 Q7b [3] Conducting tissue splits into **xylem** and **1 = phloem**. Xylem: xylem fibre, **3 = tracheid**, vessel, **5 = xylem parenchyma**. Phloem: **2 = sieve element (sieve tube)**, companion cell, **4 = phloem fibre**, **6 = phloem parenchyma** (order of 2, 4, 6 may vary) (check).

P23 Q9a [3] / Q9c [2] (a) **X: weedkiller (herbicide) — kills broad-leaved weeds**; **Y: ethylene**; **Z: stimulates stem elongation and seed germination / breaks dormancy**. (c) (i) **Translocation** (food transport in phloem). (ii) **Source:** leaf (or storage organ); **Sink:** root, fruit, growing bud, seed or tuber.

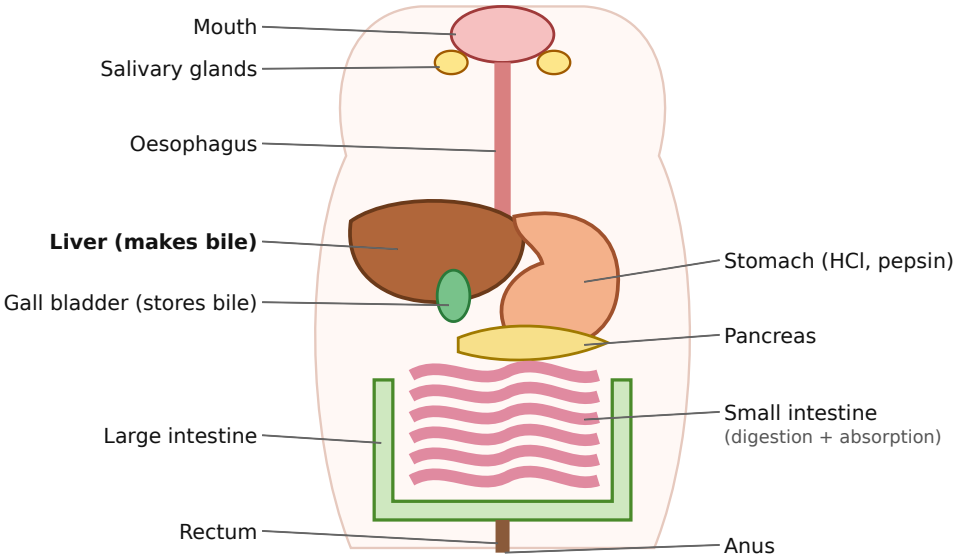
P23 Q7d(i) [1] & P23 Q1a(xvii,xviii) Q7d(i): sugars made in the leaves **cannot pass down** the cut phloem, so they **collect above the girdle** and the tissue swells. **Q1a(xvii)** (dam): consequences are loss of biodiversity and depletion of habitats → **D** by reasoning, but option **C** is printed in bold in the paper copy — **(check with your teacher)**. **Q1a(xviii):** more photosynthesis makes more sugar to export, so **A (increases food transport)** by reasoning; option C is also in bold in the paper copy **(check)**.

4 · HUMAN DIGESTION & NUTRITION

P25 Q1a(v,ix,xvii,xxiii),c(ii),Q2a,Q9a · P24 Q1a(vii),b(ii),Q4a–d,Q9b · P23 Q1a(vii,viii),b(iv),Q2b,Q4a(ii),Q6b,d,Q8b

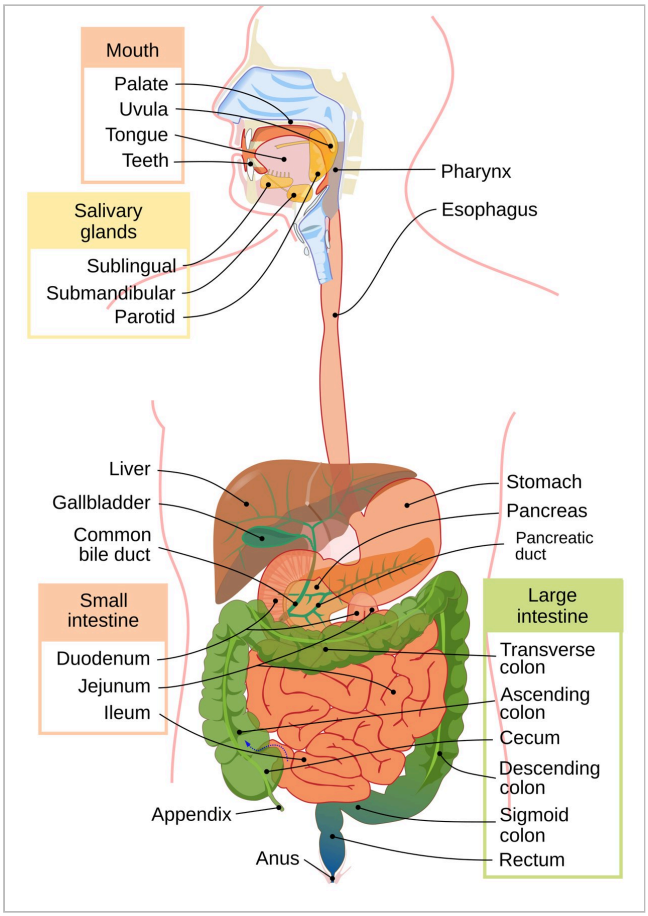
The digestive system

The human digestive system



P25 Q2a: A liver, B stomach, C pancreas, D small intestine

Human digestive system with the glands: salivary glands, liver (A), gall bladder, stomach (B), pancreas, small and large intestine.



Human digestive system (labelled diagram).
Image: Mariana Ruiz, Jmarchn, Public domain, Wikimedia Commons.

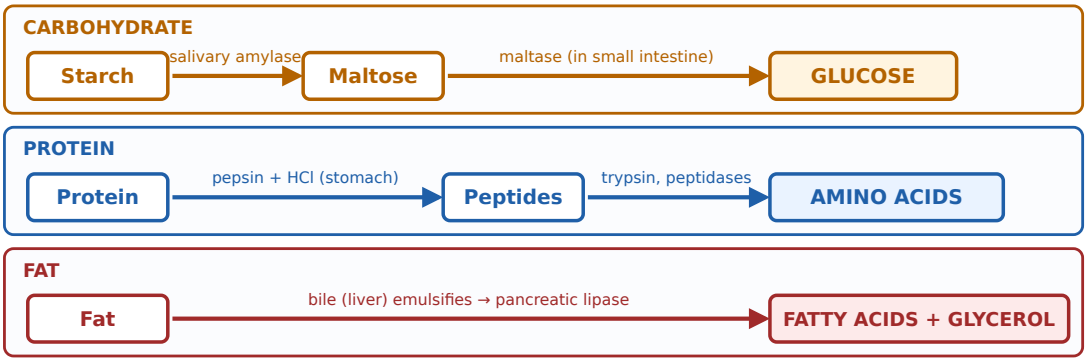
- **Path of food:** mouth → pharynx → **oesophagus** → **stomach** → **small intestine** (duodenum, jejunum, ileum) → **large intestine** → rectum → anus. **Glands:** salivary glands, **liver** (makes bile; the gall bladder stores it), **pancreas**.
- 2024 figure: **A = liver**, **B = stomach** (check). 2025 figure: **A = liver**, B = stomach, **C = pancreas**, **D = small intestine** P25 Q2a.

Who digests what

Organ	Juice / enzyme	Action
Mouth	**Salivary amylase**	starch → maltose (only **carbohydrate** starts here); teeth chew; mucus lubricates

Oesophagus	—	peristalsis pushes food (no digestion)
Stomach	**HCl**, **pepsin**, mucus, gastric lipase	HCl **kills germs, gives acid medium and activates pepsinogen → pepsin** P23 Q1b(iv); pepsin: protein → peptides; **mucus protects the stomach wall** from acid and pepsin P24 Q4a(iii); gastric lipase breaks fats P23 Q4a(ii)
Liver / gall bladder	**Bile** (no enzymes)	**Emulsifies fats** (breaks big droplets into tiny ones: more surface for lipase) and makes the food **alkaline** P24 Q4a(ii) P23 Q1a(vii): bile juice
Pancreas	Amylase, **trypsin**, lipase	starch → maltose; proteins → peptides; fats → fatty acids + glycerol
Small intestine	Maltase, peptidases, lipase	maltose → **glucose**; peptides → **amino acids**; **absorption** through villi
Large intestine	—	absorbs **water and salts**; forms faeces

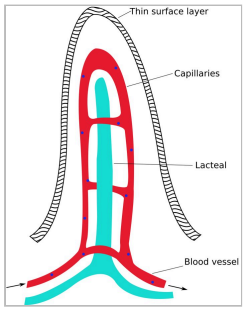
Digestion: which enzyme, where, end product



Absorption of the end products happens through the villi of the small intestine; water is absorbed in the large intestine.

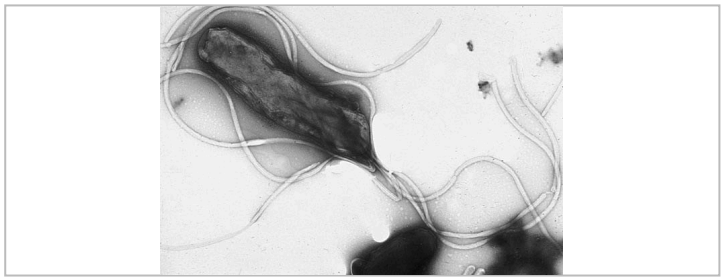
Digestion of carbohydrates, proteins and fats: enzymes and end products.

- **Carbohydrate** (rice, bread): starch → maltose → **glucose** (the final product that cells absorb) P24 Q1b(ii). **Protein** (beef, beans): → peptides → **amino acids**. **Fat:** → **fatty acids + glycerol**.
- **Small intestine adaptations (part D):** very **long and coiled**; wall folded into **villi** with **microvilli** → huge surface area; **thin (one-cell) lining**; rich **capillary network and lacteal** to carry absorbed food P25 Q2a(ii) P25 Q1c(ii): TRUE.
- Pancreatic disease (chronic pancreatitis) → **reduced digestion of carbohydrates, proteins and fats → B** P25 Q1a(v). Gastritis (less gastric juice) → **decreased breakdown of proteins in the stomach → C** P25 Q1a(xxiii).



Villi in the small intestine: capillaries and a lacteal in each villus.
Image: Snow93, Public domain, Wikimedia Commons.

Helicobacter pylori (H. pylori)



Helicobacter pylori bacteria on the stomach lining.
Image: Yutaka Tsutsumi, M.D. Professor Departme..., Copyrighted free use, Wikimedia Commons.

- A **spiral bacterium living in the stomach lining**; it survives the acid by making **urease** (changes urea to ammonia). It causes **gastritis, stomach (peptic) ulcers** and raises the risk of **stomach cancer** P25 Q1a(ix): C P24 Q4d(i).
- **Why a concern in Bhutan:** high rates of infection and of **gastric cancer**; spreads through **contaminated food/water and shared utensils**; the Ministry of Health uses **triple therapy for 14 days** and endoscopy for those 40 and above P23 Q6b(i).
- **Diagnosis: urea breath test** or **stool antigen test** (also blood test, endoscopy with biopsy) P23 Q6b(ii) P24 Q4d(iii). **Screening** lets us treat early, **preventing ulcers and cancer and stopping spread** P24 Q4d(ii).

Healthy digestion & diet

- **Eat mindfully and chew thoroughly** P24 Q1a(vii): D. **Regular exercise** improves blood flow and **peristalsis**. A **regular eating schedule** trains the gut to release juices on time, so digestion and absorption are steady P24 Q9b.
- **Constipation — prevent by:** eating **fibre** (vegetables, fruits, whole grains), **drinking plenty of water**, **regular**

exercise, regular toilet habits, **less junk/processed food** P23 Q8b.

- **Processed food** contains additives, excess salt, sugar and fat → **lifestyle diseases** (obesity, diabetes, high blood pressure, kidney disease) → **B** P25 Q1a(xvii).

Solved questions

P25 Q2a [3] (i) **A = liver**; **C = pancreas**. (ii) Structural adaptation of D (small intestine): **inner wall has villi and microvilli** to increase the surface area for absorption (or: long, thin walled, rich blood supply). (iii) HCl in the stomach (B): **kills bacteria in food, provides the acid medium for pepsin and activates pepsinogen into pepsin**.

P24 Q4 [8] (a)(i) **Part A = liver**, **Part B = stomach** (check). (ii) **Bile emulsifies fats** into small droplets and makes the food alkaline. (iii) **Mucus protects the stomach lining** from the acid and pepsin. (b)(i) **Rice** (starch begins digestion by salivary amylase). (ii) **Beef (and beans)** — protein is absorbed as amino acids. (c)(i) **Mr B** takes **more carbohydrate at every meal**; the excess is converted to **glycogen and then stored as fat**, so he gains more weight. (ii) **No** — excess carbohydrate causes obesity, diabetes, tooth decay and heart disease; eat a balanced diet. (d) See H. pylori notes above.

P24 Q9b [4] (i) Exercise increases **peristalsis and blood flow** to the gut, so food moves and is digested faster. (ii) A **gallstone blocks the bile duct**, so **less bile reaches the small intestine** — fat is not emulsified, so poorly digested; pain and fatty stools. (iii) **Comment:** chewing gum makes saliva and gastric juice flow **with no food**, so too much can cause acid, gastritis and bloating (swallowed air); a little sugar-free gum after meals is acceptable. (iv) Regular meals let the gut **release enzymes and bile at set times**, give time to digest and absorb fully and prevent overeating and acidity.

P23 Q2b [2] / Q4a(ii) [1] Flow chart: mouth (not digested) → stomach (digested) → small intestine (digested) → villi (absorbed). (i) Food = **protein** (digestion begins in the stomach). (ii) Final product: **amino acids**. (iii) Enzymes: **pepsin** (stomach), **trypsin** and **peptidases** (small intestine). Q4a(ii): fat-digesting enzyme in the stomach = **gastric lipase**.

P23 Q6d [2] & Q8b [2] Q6d: **Partly agree.** Good: traditional diet of **red rice, buckwheat, vegetables, legumes, little processed food**; **but** many eat **too much salt, butter/cheese and chilli, and more instant noodles, sweets and fizzy drinks** today, so improvement is needed. Q8b: fibre-rich food, drink plenty of water, regular exercise, regular toilet routine, avoid junk food (any four).

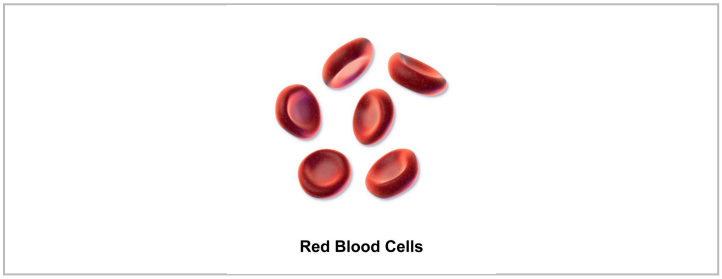
P25 Q9a [2] I **agree**. (1) Students will get **healthy meals** (fruits, vegetables, grains) instead of high salt, sugar and fat foods, lowering the risk of obesity, diabetes and kidney disease. (2) It builds **lifelong healthy habits** and improves **concentration and health** of students.

5 · BLOOD & BLOOD GROUPS

P25 Q1a(vi),b(ii),Q6c,Q8b,Q9b · P24 Q1a(viii),d(i,ii),Q6a,b · P23 Q1a(ix),b(v),Q3c,Q4a(iii)

Components of blood

Component	Main facts
Plasma (about 55%)	Liquid part: **water, proteins, salts, glucose, hormones, wastes**; carries dissolved substances P24 Q1d(i): a
Red blood cells (RBC, erythrocytes)	**Haemoglobin** carries **oxygen** (and some CO_2) P25 Q1a(vi): C P24 Q1d(ii): c; biconcave, **no nucleus** → more space for haemoglobin P25 Q9b(i)
White blood cells (WBC, leucocytes)	**Immunity**: defend against germs; **phagocytosis** = engulfing pathogens P23 Q1a(ix): C P23 Q4a(iii); types: lymphocytes, monocytes, neutrophils, eosinophils, basophils
Platelets (thrombocytes)	Cell fragments that help **clot the blood** and stop bleeding P25 Q8b

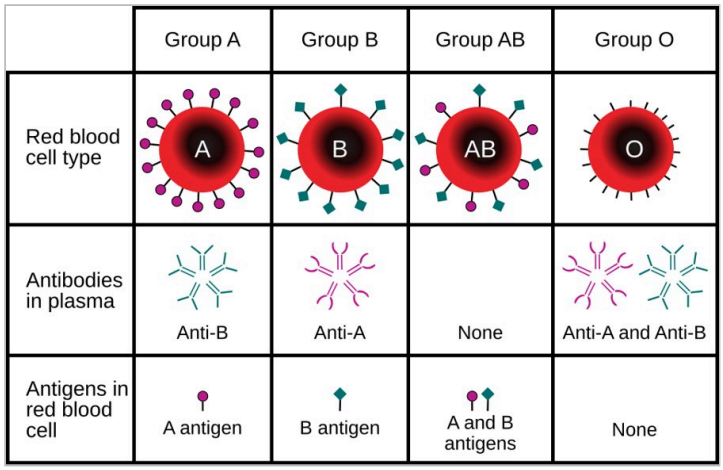


Red blood cells: biconcave discs with no nucleus (haemoglobin carries O_2).
Image: BruceBlaus. When using this image in ext..., CC BY 3.0, Wikimedia Commons.

- **Blood report (2024):** monocytes 12% (normal 1–9%) are **raised** → the body is fighting an **infection** → **B** P24 Q1a(viii) (check).

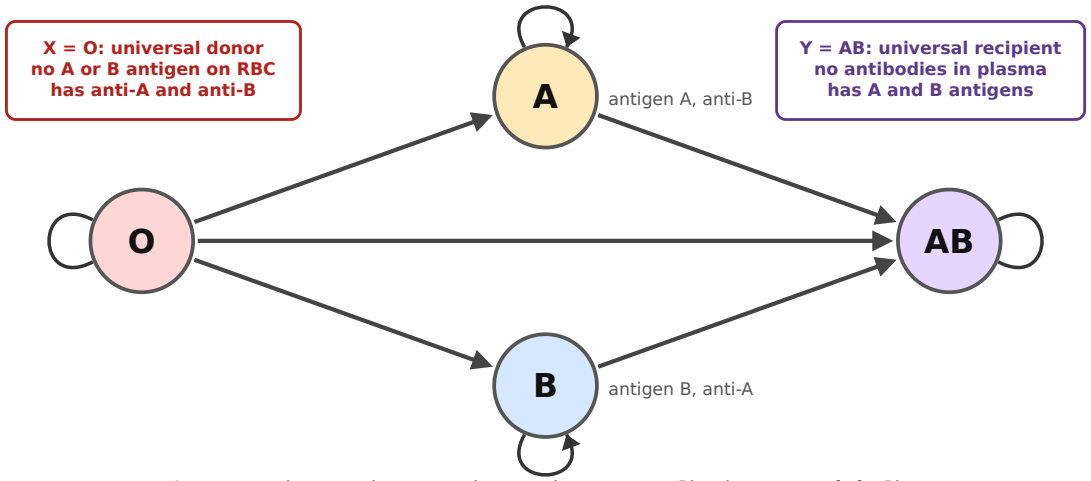
ABO and Rh blood groups

Group	Antigen on RBC	Antibody in plasma	Can give to	Can receive from
A	A	**anti-B**	A, AB	A, O
B	B	anti-A	B, AB	B, O
AB	A and B	**none**	AB only	**all** (universal recipient)
O	**none**	anti-A and anti-B	**all** (universal donor)	O only



Antigens on the red cells and antibodies in the plasma for each ABO group.
Image: InvictaHOG, Public domain, Wikimedia Commons.

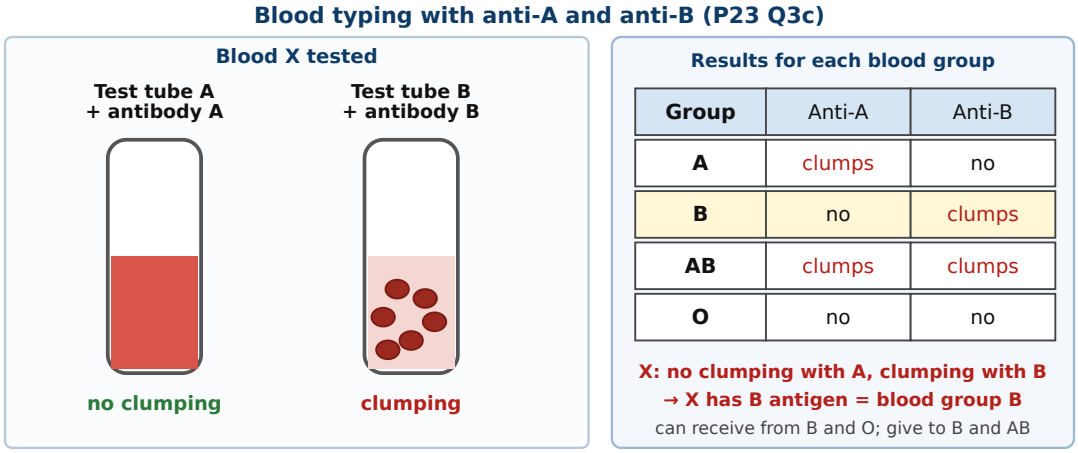
Who can give blood to whom (ABO system)



Arrow = can donate to; loop = can donate to its own group. Rh− donors are safe for Rh− and Rh+; Rh+ blood must only go to Rh+.

Donor–recipient chart: O is the universal donor (X), AB the universal recipient (Y) — P25 Q6c.

- A person of group A has **no anti-A** antibody in the plasma; it has **anti-B** P25 Q1b(ii). Wrong blood causes **agglutination (clumping/coagulation)**: the recipient's antibody binds the donor's antigen.
- **Rh factor:** Rh^+ has the antigen, Rh^- does not. An Rh^- person makes anti-Rh only after exposure to Rh^+ blood. **O^+ receives only from O^+ and O^-.**



Blood typing test (P23 Q3c): clumping with anti-B means the blood is group B.

Solved questions

P25 Q6c [3] (i) The donor with arrows to **all groups** is **X = O** (universal donor; no A or B antigens); the group that receives from all is **Y = AB**. (ii) **AB is the universal recipient** because its plasma has **no anti-A or anti-B antibodies**, so it will not clump any donor red cells.

P25 Q9b [2] & Q8b [1] (i) RBCs **lack a nucleus** so there is **more room for haemoglobin** to carry oxygen. (ii) Group A blood has **A antigen**; group B plasma has **anti-A** antibody, which would **clump** the donor's red cells. Q8b: **platelets help blood to clot** and seal a wound.

P24 Q6a [2] & Q6b [2] (a) Group **O** blood has **both anti-A and anti-B antibodies** in the plasma, so it would **agglutinate (clump) red cells carrying A or B antigens** from A, B or AB donors; only blood without A and B antigens (group O, and matching Rh) is safe. (b) The donor must have **no anti-A antibody, no B antigen and no Rh factor**, because the patient has **A antigen** (would be attacked by anti-A), **no B antigen** (so has anti-B and would attack B antigen) and is **Rh−** (would make anti-Rh). (i) Patient = **A negative (A^-)**. (ii) Antibody A in the donated blood would **attack the patient's A-antigen red cells**: **agglutination**, blocked vessels, possible kidney failure and death.

P23 Q3c [4] Test tubes: no clumping with **anti-A**, clumping with **anti-B**. (i) X belongs to **group B** (it has B antigen only). (ii) Recipients for B blood: **B and AB**. (iii) **Coagulation (clumping)** occurred in tube B because the **anti-B antibody in the reagent reacts with the B antigen on the red cells**, joining them together (agglutination).

P23 Q1b(v) / P24 Q1d A donor who can give to both A and O patients: donor **O** (universal donor) — but a patient of O can take only O blood, so the donor is **O**. P24 matching: i **a** plasma; ii **c** red blood cells.

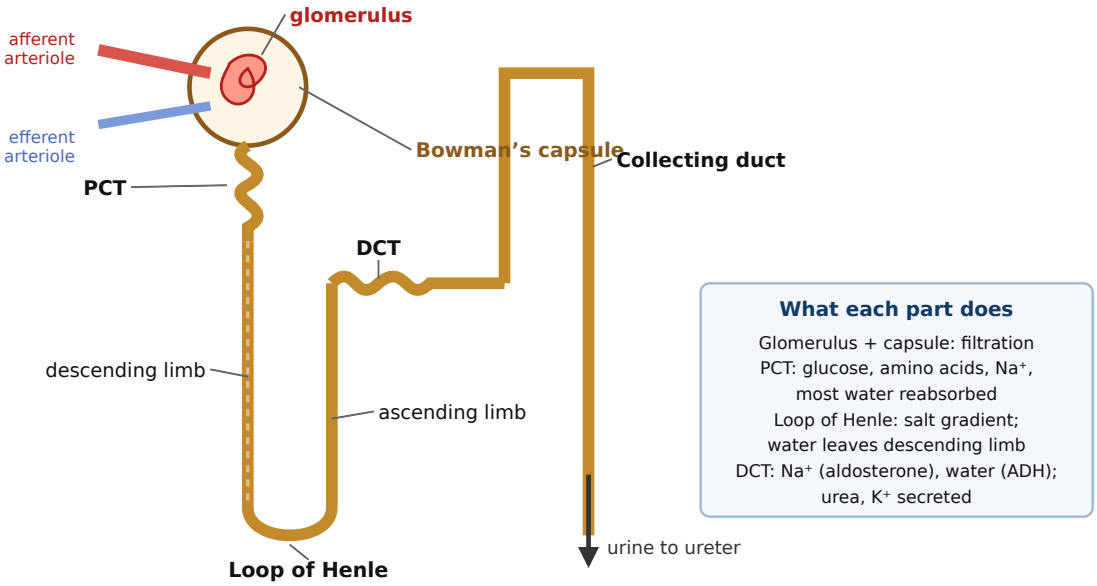
6 · EXCRETION: THE KIDNEY & NEPHRON

P25 Q1a(viii),b(iii),Q7b,c · P24 Q1a(x),d(iii–v),Q6c · P23 Q1a(xii),c(v),Q5a,c

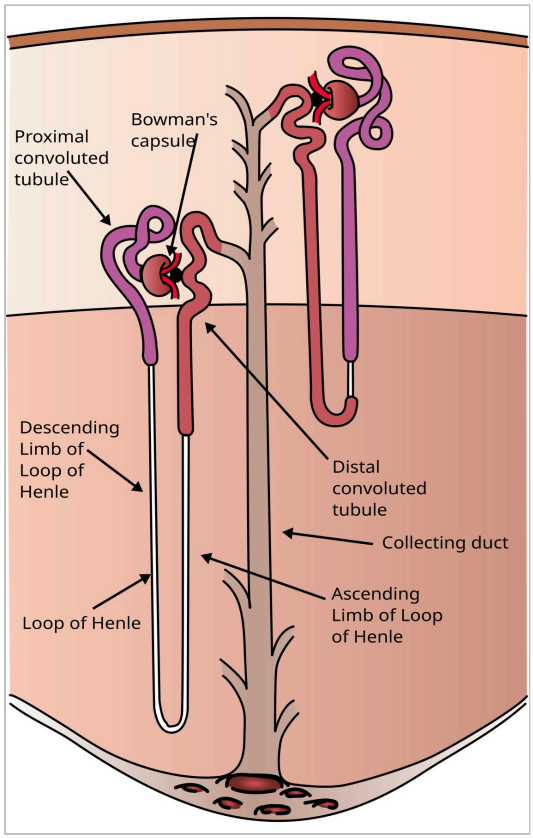
Key ideas

- **Excretion** = removal of metabolic wastes: **urea, uric acid, creatinine, excess water and salts** (kidneys → urine; skin → sweat; lungs → CO_2). Urinary system: **kidney → ureter → urinary bladder → urethra**.
- The **nephron** is the structural and functional unit of the kidney (about one million in each kidney). **Urine is formed from blood by the nephron → TRUE** P23 Q1c(v).

The nephron (draw and label)



Nephron (draw this for 3 marks): Bowman's capsule with glomerulus, PCT, loop of Henle, DCT and collecting duct.



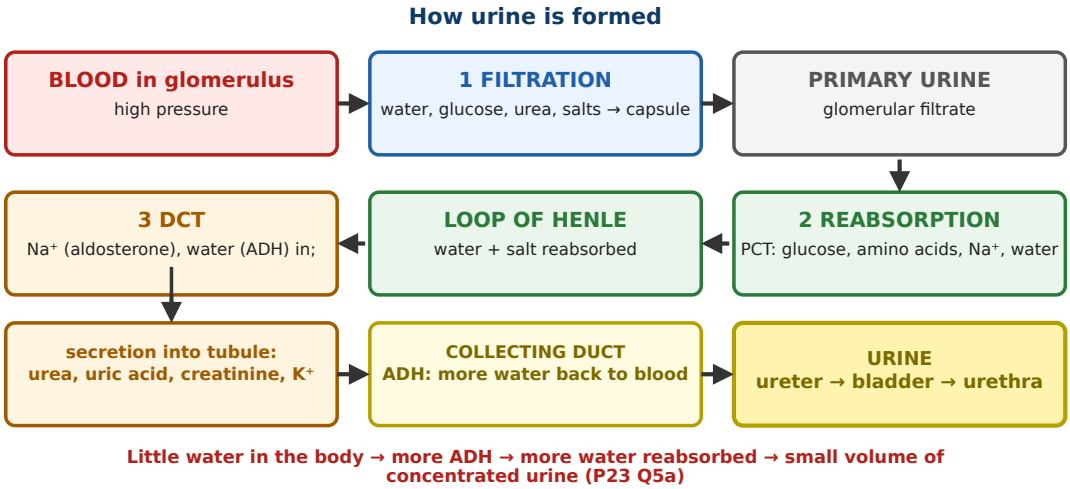
Structure of a nephron and its blood supply.
Image: Artwork by Holly Fischer, CC BY 3.0, Wikimedia Commons.

Parts of the nephron

Part	Job
Glomerulus (capillary knot)	High-pressure **filtration** of blood
Bowman's capsule	Cup that collects the filtrate (**primary urine**)
Proximal convoluted tubule (PCT)	Reabsorbs **all glucose and amino acids**, most **Na^+ (active)** and water (passive) P24 Q1d(iii): e

Loop of Henle	Makes a **salt gradient** in the kidney so water can be reabsorbed: the **descending limb is permeable to water** P25 Q1b(iii): descending ; ascending limb takes out salts P24 Q1d(iv): f
Distal convoluted tubule (DCT)	Fine control: **Na^+ (aldosterone)** and **water (ADH)** reabsorbed; **secretion** of urea, uric acid, creatinine, K^+
Collecting duct	Takes urine to the pelvis; **water reabsorbed under ADH** P24 Q1d(v): h (check — g if your text uses DCT)

How urine is formed (3 steps)



Urine formation: filtration, selective reabsorption and secretion (flow chart of P23 Q5a).

- **1 Ultrafiltration** in the glomerulus: water, glucose, amino acids, urea and salts pass into the capsule; **blood cells and big proteins stay in the blood**.
- **2 Selective reabsorption:** useful substances (all glucose, most water, salts) return to the blood. The composition graph shows **glucose in the filtrate but none in urine → glucose is reabsorbed** P23 Q1a(xii): C. **Urea is more concentrated in urine** (not reabsorbed).
- **3 Tubular secretion:** urea, uric acid, creatinine and excess ions are added from the blood into the tubule. Urine then flows through the collecting duct, ureter and bladder.
- **Hormones: aldosterone** → reabsorption of **sodium** in the DCT; **ADH (antidiuretic hormone)** → reabsorption of **water** in the DCT / collecting duct P23 Q5a(i). When the body has **little water**, more ADH is released, more water is reabsorbed, so a **small volume of concentrated urine** is passed P23 Q5a(ii).

Kidney health

- **Low physical activity → obesity and hypertension →** damage to the kidney → **C** P25 Q1a(viii). **Healthy lifestyle choices can prevent or delay kidney failure** in many cases → **D** P24 Q1a(x).
- **Unhealthy practices that harm the kidney: too much salt** (salted butter tea, pickles, processed food), **too much sugar and red meat**, **drinking too little water**, **alcohol and smoking**, overuse of painkillers, obesity and high blood pressure, diabetes.
- Treatment of kidney failure: **dialysis** or kidney **transplant**.

Solved questions

P25 Q7c [3] Draw a **long, tube-like nephron** as in the figure and label: **Bowman's capsule** (cup) with the **glomerulus** inside it, then PCT, **loop of Henle** (U-shaped), **distal convoluted tubule**, and the **collecting duct**. Use ruled label lines and a title. (1 mark each for correct drawing and labels.)

P25 Q7b [1] Two unhealthy dietary practices in our community: **(1) eating too much salt (salty suja, pickles, instant noodles); (2) eating a lot of red meat/fatty and sugary processed foods, and drinking too little water**. (Also alcohol.)

P24 Q1d(iii–v) / Q6c [2] Match: iii **e** (proximal tubule); iv **f** (loop of Henle); v **h** (collecting duct) (check). Q6c: **Y = proximal convoluted tubule (PCT)**; **Z = distal convoluted tubule (DCT)**; X = **glomerulus**. If X is damaged, **blood is not filtered**: little or no filtrate (urine) forms, **wastes build up** in the blood and proteins/blood cells may leak into urine → kidney failure.

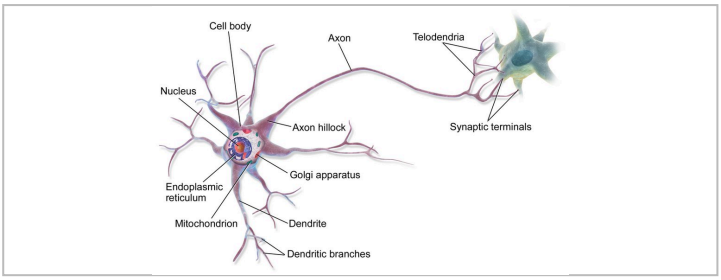
P23 Q5a [4] & Q5c [1] (i) A: **aldosterone** (Na^+ reabsorption); B: **ADH / antidiuretic hormone** (water reabsorption). (ii) When water is short, **more ADH** is secreted, the DCT/collecting duct **reabsorb more water** into the blood, so **less water goes out** as a small volume of concentrated urine. Q5c: **too much salt/sugar and little water; painkillers/alcohol; obesity or untreated high blood pressure** (any two).

7 · NERVOUS SYSTEM, COORDINATION & STEROIDS

P25 Q1a(xxi),c(iii),Q3a,Q9c · P24 Q9a · P23 Q1a(x,xi),Q4b,Q7a

Key ideas

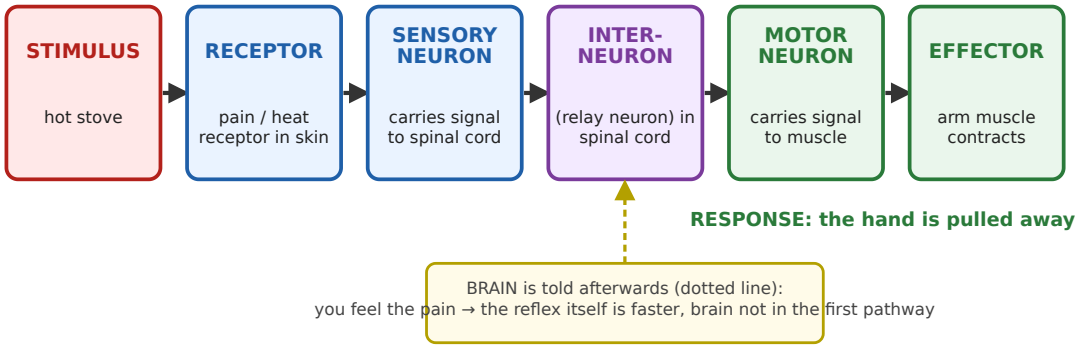
- The **nervous system** detects changes (**stimuli**) and produces a quick **response** using electrical signals. **CNS** = brain + spinal cord; **PNS** = nerves.
- **Neurons: sensory neuron** (receptor → CNS), **interneuron / relay neuron** (inside CNS), **motor neuron** (CNS → effector). Parts of a neuron: dendrites, cell body (with nucleus), **axon** (with myelin sheath), axon terminals.
- **Brain: cerebrum** (thinking, memory, voluntary actions), **cerebellum** (balance), **medulla** (heartbeat, breathing).
- **Receptors** (sense organs: eye, ear, skin, nose, tongue) detect stimuli; **effectors** (muscles, glands) respond.



A motor neuron with its parts labelled.
Image: BruceBlaus, CC BY 3.0, Wikimedia Commons.

The reflex arc

Reflex arc: pulling the hand from a hot stove



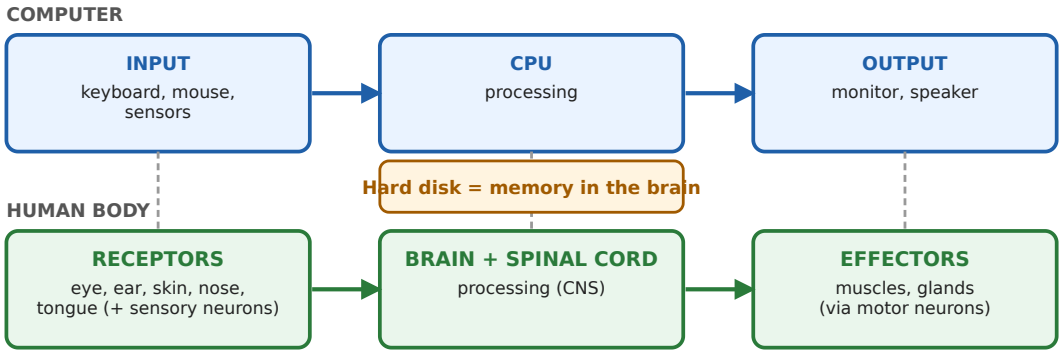
Receptor → sensory neuron → interneuron → motor neuron → effector (P25 Q1a(xxi): option C)

Reflex arc: hot stove → receptor → sensory neuron → interneuron (spinal cord) → motor neuron → effector (muscle).

- Karma touches a hot stove and pulls his hand away: **Receptor → Sensory neuron → Interneuron → Motor neuron → Effector → C** P25 Q1a(xxi).
- A **natural (spinal) reflex** is processed in the **spinal cord**, so it is very fast; the **brain is not in the initial pathway** (it is told afterwards) → "natural reflexes involve the brain in the initial response" is **FALSE** P25 Q1c(iii).

Computer and nervous system

The nervous system works like a computer



Both use electrical signals. A computer cannot make its own decisions; it follows a program.

The nervous system works like a computer: input → processing → output, with storage.

Computer	Nervous system
Keyboard / sensors / mouse (input)	**Receptors / sense organs** and sensory neurons
CPU (processing)	**Brain and spinal cord** (CNS)
Monitor / speaker (output)	**Effectors**: muscles and glands (via motor neurons)
Hard disk (storage)	**Memory** stored in the brain (cerebrum)

- **Similarities:** both take inputs from sensors, process them, give outputs, and **depend on electrical signals**. **Not true:** "both can make their own decision" — a computer only follows its program → **C** P23 Q1a(x).

Steroids

- **Anabolic steroids** are synthetic hormones like testosterone that **build muscle and body mass**. They are **banned in sport** because they give an **unfair advantage** and are **harmful** P25 Q3a.

- **Physical side effects:** infertility, shrunken testes, **irregular periods**, spotting, acne, liver damage, high blood pressure, heart disease, deep voice and facial hair in women. **Muscle development is the intended effect, not a side effect → D** P23 Q1a(xi).
- **Psychological side effects: mood swings, aggression ("roid rage"), depression, anxiety, irritability, addiction/dependence, paranoia** P23 Q7a.

Solved questions

P25 Q9c [2] 1) **CPU** = **brain and spinal cord (CNS)** which process information; 2) **Keyboard** = **receptors / sense organs** (input); 3) **Monitor** = **effectors (muscles, glands)** that give the response; 4) **Hard disk** = **memory in the brain** (storage).

P25 Q3a [2] Athletes may not use anabolic steroids because: (1) it gives an **unfair advantage** over athletes who compete honestly; (2) it has **serious side effects** (heart and liver damage, infertility, aggression and depression).

P23 Q4b [2] A computer takes **input** from the keyboard/sensors, the **CPU processes** it and gives **output** on the screen/speaker, with **storage**. In the same way **receptors** collect stimuli, send **electrical signals** by sensory neurons to the **brain/spinal cord** (processing), which sends signals by motor neurons to **effectors** (response); memory is stored in the brain.

P23 Q7a [2] Four psychological side effects of excessive steroid use: **mood swings, aggression, depression, anxiety/irritability** (also addiction, paranoia).

P24 Q9a [6] (i) The receptor that works like a **keyboard** is the **touch receptor of the skin (or any sense organ)**: it **receives a stimulus and converts it into an electrical signal** sent to the brain, as the keyboard turns key presses into signals for the CPU. (ii) A machine **sensor** (light, heat, sound, pressure sensor) **detects a change and converts it to an electrical signal** for processing, exactly like a receptor. (iii) Limitation: receptors cover only a **limited range** (we cannot see UV or hear very high sounds) and can be **damaged/tired**; sensors need **power**, can fail or be inaccurate. (iv) They help create **robots, medical devices, artificial limbs, cameras and smart machines** and help scientists understand the body.

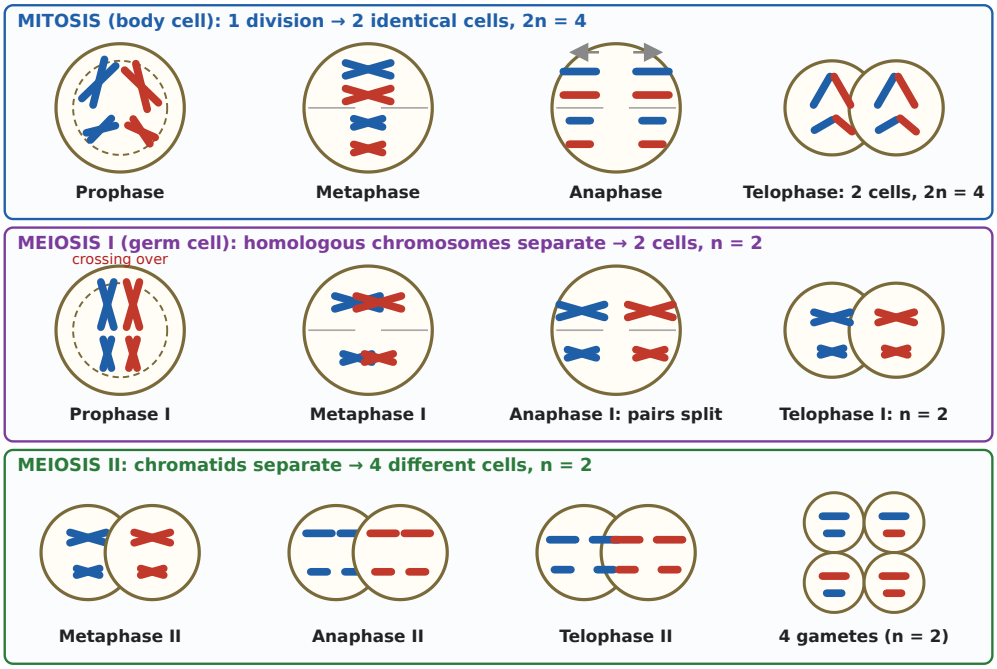
8 · CELL DIVISION & HEREDITY

P25 Q1a(xi,xii,xix,xxv),b(ii),c(v),d(v),Q2c,Q4d,e,Q5e,Q7e,Q8c,Q9d · P24 Q1a(xv–xx),b(v),Q7a,Q8a · P23 Q1a(xix–xxii),d(i,ii),Q4c,Q5e,Q7c,Q9b,e

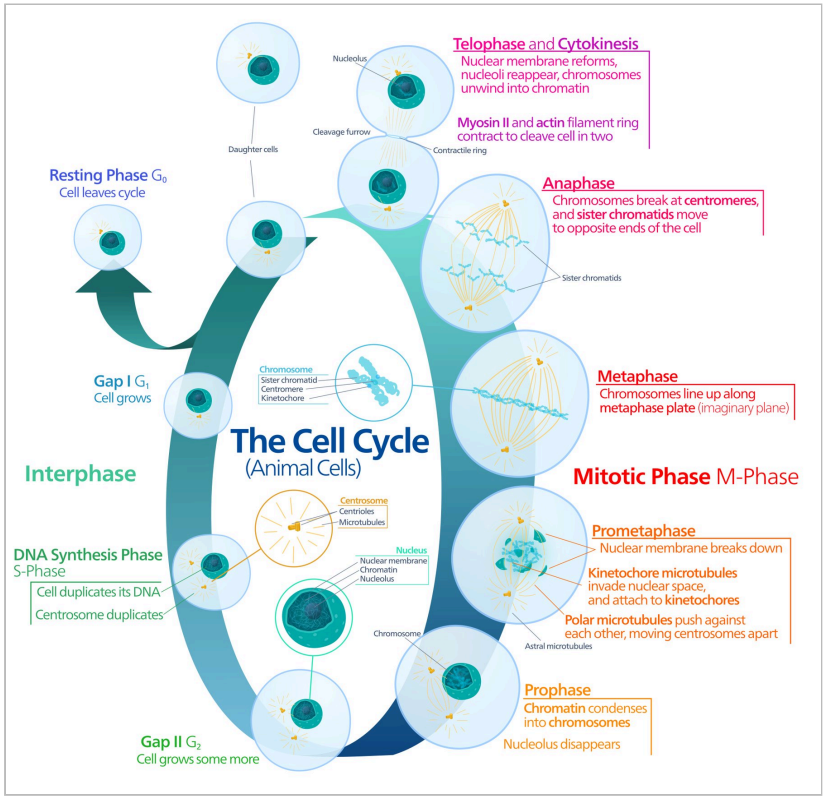
Mitosis vs meiosis

	Mitosis	Meiosis
Cells	**Somatic (body)** cells	**Germ cells** (in testes / ovaries)
Divisions	One	**Two** (meiosis I and II)
Daughter cells	**2**, **genetically identical** to the mother cell	**4**, **genetically different** gametes
Chromosomes	Same number (2n → 2n)	**Halved** (2n → n): **reduction division**
Crossing over	No	**Yes** (prophase I)
Purpose	**Growth, repair, replacement** (a cut heals by dividing nearby cells P23 Q1a(xx): D; goes on in adults P23 Q9b(iii))	**Gametes** (sperm, eggs); variation

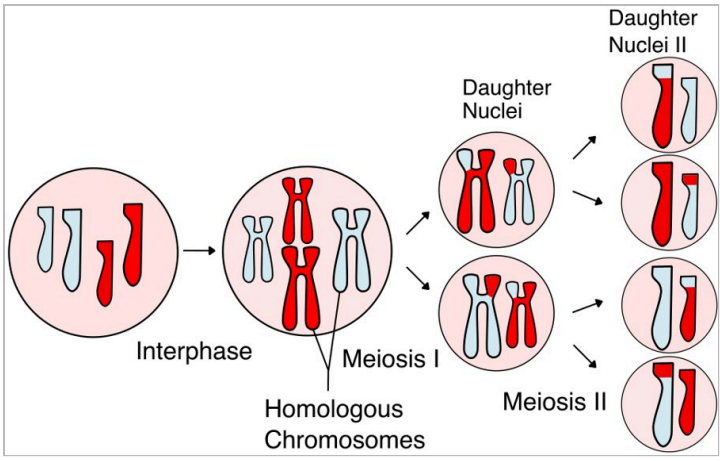
Mitosis and meiosis for a cell with 4 chromosomes (2n = 4)



Stages of mitosis and of meiosis for a cell with 4 chromosomes (2n = 4).



The animal cell cycle: interphase and the stages of mitosis.
Image: Kelvinsong, CC0, Wikimedia Commons.



Meiosis I and II: homologous chromosomes separate, then chromatids separate.

Image: Rdbickel, CC BY-SA 4.0, Wikimedia Commons.

- **Mitosis** uses **somatic cells** to create **genetically identical** cells → **D** P25 Q1a(xix). **Meiosis I** is called **reduction division** because the **homologous chromosomes separate**, so the number falls from 2n to n P25 Q4e.
- **Anaphase** (mitosis / meiosis II): the **sister chromatids separate and move to opposite poles** as spindle fibres shorten → diagram **C** (elongated cell with chromatids at the poles) P24 Q1a(xvi) (check). The figure with chromosomes (each of two chromatids) moving to the poles = **Anaphase I → C** P25 Q1a(xi) (check).

Crossing over in prophase I



Crossing over, independent assortment and random fertilisation create variation (P23 Q1a(xix): answer B).

Crossing over in prophase I: non-sister chromatids exchange segments (answer B in P23 Q1a(xix)).

- **Crossing over** (exchange of segments between homologous chromosomes), **independent assortment** of chromosomes and **random fertilisation** all make offspring different. After the crossing-over shown in P23 Q1a(xix), **each chromosome carries a piece of the other → B**.
- Meiosis **increases genetic variation**, mitosis does not → **TRUE** P25 Q1c(v).

Mendel's laws & vocabulary

- **Gene** = unit of heredity; **allele** = form of a gene. **Homozygous** (same alleles: TT, tt), **heterozygous** (different alleles: Tt) P24 Q1b(v). **Genotype** = alleles; **phenotype** = visible trait.
- **Law of dominance:** when two contrasting alleles are present, the **dominant allele masks (expresses over)** the recessive one P25 Q1a(xii): B P23 Q9e. A **recessive allele expresses only in the homozygous form** P25 Q1d(v): e. **Law of segregation:** the two alleles of a gene separate when gametes form.
- Dominant parent × recessive parent giving **all dominant offspring** → the dominant parent is **homozygous**
- **Chromosomes in a cell stuck at prophase I** (egg in a girl): **46 chromosomes and 92 chromatids** (DNA already copied) → **D** P24 Q1a(xv).
- **Haploid vs diploid:** haploid (n, gametes) has **one set** of chromosomes, no pairs; diploid (2n, body cells) has **two sets** in **homologous pairs** P25 Q5e. Body cells carry **maternal and paternal** information; sex cells carry **one set → C** P24 Q1a(xvii).

Why gametes differ: variation

dominant → B P24 Q1a(xx). Homozygous red × homozygous white (red dominant) → **100% red → B** P25 Q1a(xxv).

Punnett squares used in the papers

Tt × tt (P25 Q9d)

	T	t
t	Tt	tt
t	Tt	tt

1 Tt : 1 tt
1 tall : 1 dwarf

Tt × Tt

	T	t
T	TT	Tt
t	Tt	tt

1 TT : 2 Tt : 1 tt
3 tall : 1 dwarf (75% tall)

Ee × ee (P24 Q8a)

	E	e
e	Ee	ee
e	Ee	ee

1 Ee : 1 ee
1 free : 1 attached (50%)

EE × Ee (P23 Q7c)

	E	e
E	EE	Ee
E	EE	Ee

1 EE : 1 Ee
all free earlobes (100%)

Carrier ♀ × normal ♂ (P25 Q8c)

	XC	Xc
XC	XCXC	XCXc
Y	XCY	XcY

normal ♀, carrier ♀, normal ♂, colour-blind ♂ (c = defect): 75% norma

Haemophilia (P23 Q1a(xxi))

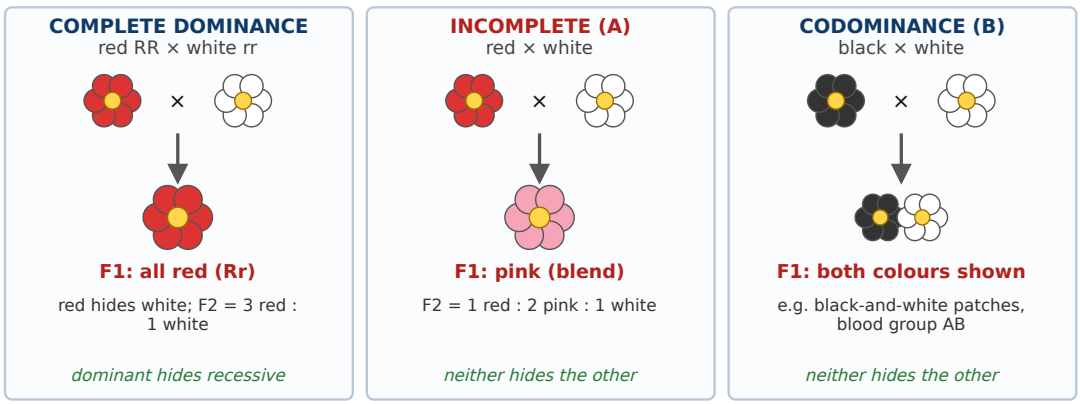
	Xh	Xh
XH	XHXh	XHXh
Y	XhY	XhY

all ♀ carriers, all ♂ haemophilic
(h = haemophilia allele)

Punnett squares for the crosses asked in the papers.

Beyond Mendel; sex-linked traits

Complete dominance, incomplete dominance and codominance



Complete dominance, incomplete dominance and codominance.

- **Incomplete dominance (A):** the heterozygote is **intermediate** (black × white → grey). **Codominance (B): both** alleles show fully (e.g. black and white patches; blood group AB). Unlike Mendelian inheritance, **no allele completely hides the other** P25 Q2c.
- **Sex:** XX = female, XY = male. **Haemophilia and colour blindness** are **recessive genes on the X chromosome** (sex-linked) P23 Q5e. They are **more common in males**, who have only one X, so a single recessive allele shows; females need two P25 Q8c(ii) P23 Q1a(xxii): C P24 Q1a(xviii): B.
- Carrier female × unaffected male: **50% of sons** are affected P24 Q1a(xix): C. A couple with **50% haemophilic children and the rest carriers** = **normal father and diseased mother** (sons affected, daughters carriers) → **B** P23 Q1a(xxi).

Solved questions

P25 Q4d [2] & Q5e [1] Diploid cell with 12 chromosomes: meiosis I halves the number (12 → 6) and meiosis II separates chromatids, so **four daughter cells, each with 6 chromosomes** (12 ÷ 2 = 6). Q5e: haploid cell = **one set** (n), diploid = **two sets in pairs** (2n).

P25 Q7e [2] & P23 Q9b [3] Two sperm from one man differ because: (1) in meiosis the homologous chromosomes **line up and are shared out at random** (independent assortment); (2) **crossing over** in prophase I swaps pieces of DNA. (P23: (i) **meiosis**; (iii) mitosis in adults **replaces dead cells and repairs wounds** (skin, blood, gut lining).)

P25 Q8c [2] (i) Mother X^CX^c × father X^CY gives X^CX^C, X^CX^c, X^CY, X^cY. **75% have normal vision** (2 daughters + 1 son; one daughter is a carrier); **25% (one son) is colour-blind**. (ii) Males have **one X**, so a single recessive allele is expressed; a female needs **two** recessive alleles.

P25 Q9d [2] Tt × tt: gametes T, t and t, t → Tt, Tt, tt, tt. (i) **Genotypic ratio 1 Tt : 1 tt**; **phenotypic ratio 1 tall : 1 dwarf**. (ii) Tt × Tt gives TT, Tt, Tt, tt → **75% tall** (3 tall : 1 dwarf).

P24 Q7a [7] Original cell (figure) has **4 chromosomes**. (i) The egg cell has **2 chromosomes** (half). (ii) Meiosis **halves the chromosome number in gametes**; fertilisation **restores 2n**, so the number stays the same in every generation. (iii) Mitosis: 1 division, **2** identical diploid cells, **no crossing over**, in body cells for growth/repair. Meiosis: **2 divisions, 4 different haploid cells**, crossing over, in germ cells. (iv) **Yes**: skin cells are body cells made by **mitosis** from the same zygote, so their DNA is **identical**.

P24 Q8a [4] Ee (free) × ee (attached). Punnett square (gametes E, e down; e, e across): Ee, Ee, ee, ee. (ii) **Phenotypic ratio 1 free : 1 attached; genotypic ratio 1 Ee : 1 ee**. (iii) The father (ee) has **attached earlobes**.

P23 Q7c [3] & Q1d(i,ii) Father homozygous dominant **EE**, mother heterozygous **Ee**: EE, EE, Ee, Ee → (ii) **100% free earlobes**; (iii) phenotypic ratio **4 free : 0 attached** (all free). Matching: i **d** mitosis; ii **c** dominant allele; iii **e** meiosis; iv **a** selective breeding; v **b** evolution.

P23 Q4c [3] The phase with chromosomes at the equator joined to spindle fibres (in two columns of pairs in meiosis I, one plate in mitosis) is **metaphase (metaphase I)** (check). The phase before it is **prophase**: draw a cell with **thickened chromosomes (in meiosis, homologues paired with crossing over), nuclear membrane disappearing, spindle forming**, and label “Prophase”.

P25 Q2c / Q1c(v) Figure A = **incomplete dominance** (grey from black × white), B = **codominance** (both colours shown). Difference from Mendelian: in Mendel’s law the heterozygote looks like the **dominant** parent; here the heterozygote shows a **blend or both traits** because neither allele is completely dominant.

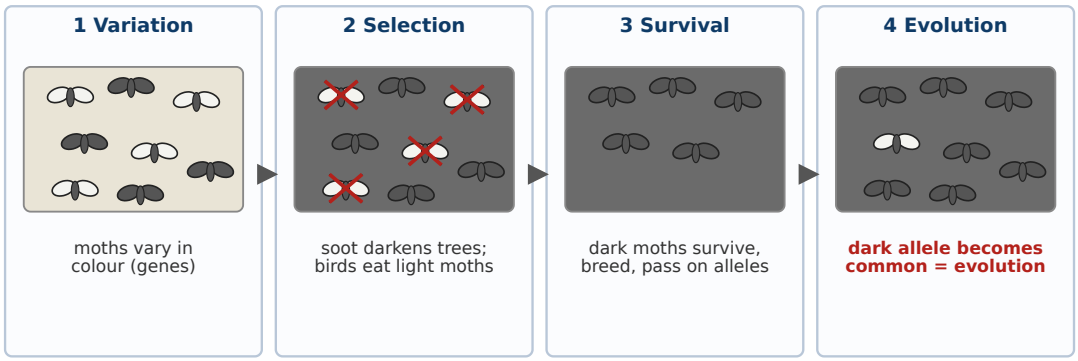
9 · EVOLUTION & BIOTECHNOLOGY

P25 Q1a(xiii–xv),b(iv,v),d(iii,iv),Q2b,d,Q3d,e,Q5b,c,Q6b,Q7d,Q8f · P24 Q1a(xxi–xxv),c(iv,v),Q3b,Q6d,Q8b · P23 Q1a(ii,xiii,xxiii–xxv),c(iii),Q2d,Q5d,Q6c,Q7b,d,Q8d

Natural selection

- **Steps:** (1) **variation** in a population; (2) **more offspring are born than can survive** (competition); (3) individuals with **favourable traits (best adapted)** survive and reproduce — **fitness** = **ability to survive and reproduce in a given environment** P24 Q1a(xxii): D; (4) they pass the **advantageous alleles** on, so their frequency **increases** over time P24 Q1a(xxiv): A.
- **Example of natural selection:** giraffes with longer necks reaching tall trees survive and breed → **D** P24 Q1a(xxi).
- **Peppered moths:** soot from factories darkened trees, so **dark moths were camouflaged and survived**; white moths were eaten. A change in the environment can result in **evolution** → **B** P23 Q1a(xxiii).
- Warming reduces snow, so **black rabbits** are better camouflaged on dark soil and thrive → **C** P25 Q1a(xiii). Mice graph: brown mice **increase**, white mice **decrease** → **natural selection** favours **brown** (camouflaged in the bushy habitat) P23 Q5d.
- **Sickle-cell trait** is common where **malaria** is endemic: carriers resist malaria, so natural selection keeps the allele P25 Q1b(v).

Natural selection: the peppered moth



Fitness = ability to survive and reproduce. Advantageous alleles increase in frequency.

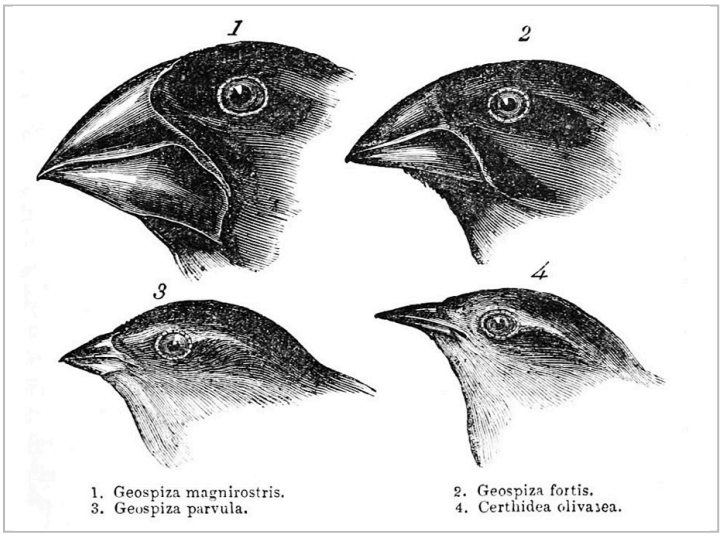
Natural selection: variation → selection → inheritance → change over time (peppered moth example).



Peppered moth, light form (typica).
Image: Chiswick Chap, CC BY-SA 3.0, Wikimedia Commons.



Peppered moth, dark form (carbonaria).
Image: Chiswick Chap, CC BY-SA 3.0, Wikimedia Commons.

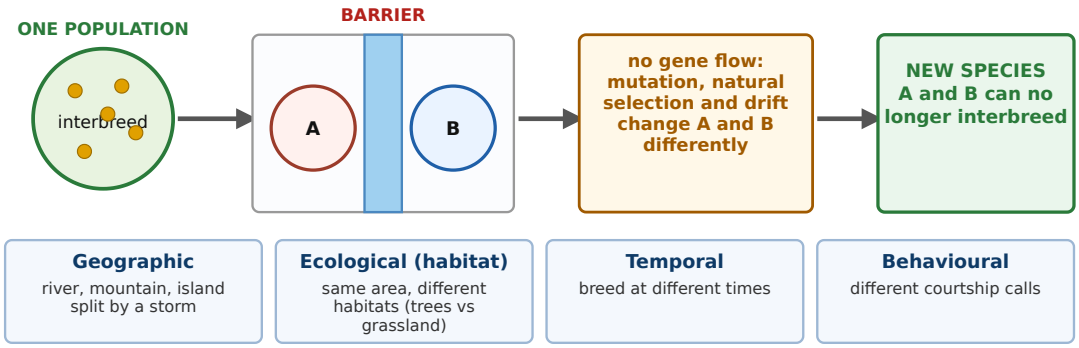


Darwin's finches: beak shape matches the food (seeds, insects) — P23 Q8d.
Image: John Gould (14.Sep.1804 - 3.Feb.1881), Public domain, Wikimedia Commons.

Isolation & speciation

Type of isolation	Meaning / example
Geographic	A **physical barrier** (island split by a storm, river, mountain) separates populations P25 Q5c: Situation B
Ecological (habitat)	Populations live in **different habitats** in the same area: lizards in different parts of a desert P25 Q5c: Situation A; squirrels in trees vs grassland P23 Q1a(xxv): B
Temporal	Breed at **different times** (seasons / hours)
Behavioural	Different **courtship** signals / calls

Isolation leads to speciation



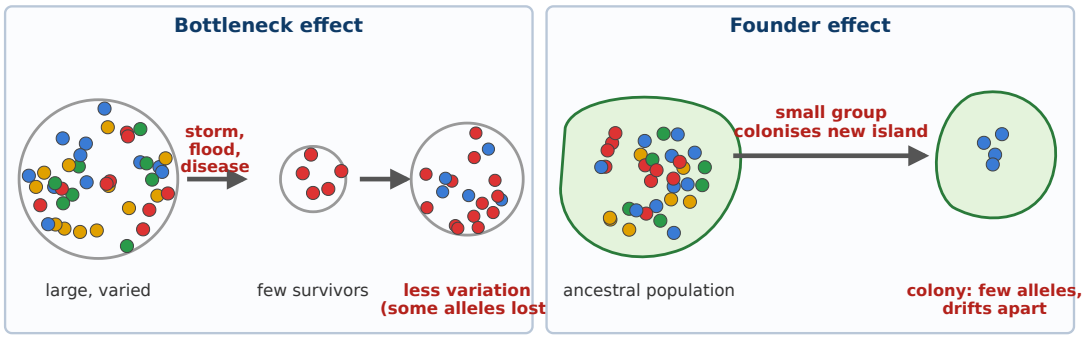
P25 Q5c: lizards in one desert, different habitats = ecological; island split by a storm = geographic

Isolation → no gene flow → differences build up → new species (speciation).

- **Speciation** = formation of a new species. Isolated populations **cannot interbreed**, so mutation, selection and drift make them different until they are **separate species**. **Mutation** gives new alleles and variation, the raw material for speciation P25 Q3d; **variation lets natural selection act, so allele frequencies change = evolution** P23 Q7b(ii), Q7d(ii).
- **Gene flow** = **exchange of alleles between populations** (migration) → **D** P25 Q1a(xv).

Genetic drift

Genetic drift: bottleneck effect and founder effect



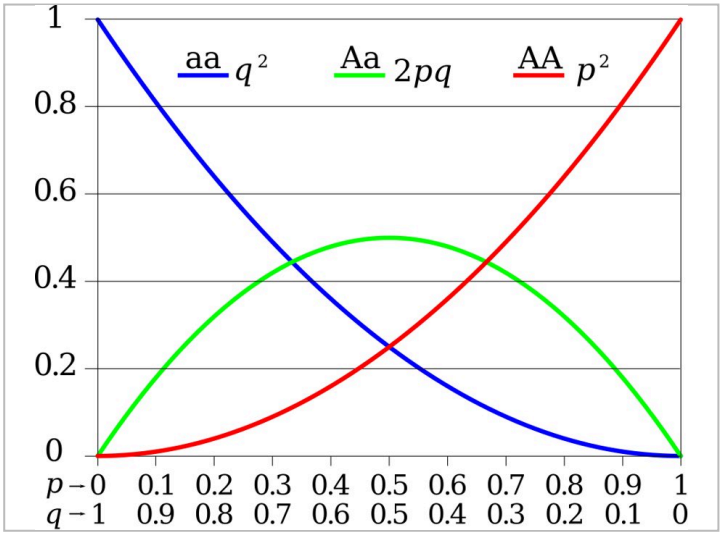
Drift = chance change in allele frequency, strongest in small populations.

Bottleneck effect and founder effect: chance changes in allele frequency in small populations.

- **Genetic drift** = **random** change in allele frequency, strongest in **small** populations. **Bottleneck effect:** a storm, flood or disease kills most of a population, so **genetic diversity falls** P25 Q1b(iv) P23 Q2d. **Founder effect:** a **small group colonises a new area** (figure: "colonisers" leave the ancestral population) P24 Q8b(i).
- **Cheetahs** have **little genetic variation**, so a disease or new predator can wipe out the population because **no individual has a different, resistant trait** P24 Q3b.

Hardy–Weinberg principle

- In a **large, randomly mating population with no mutation, migration, selection or drift**, allele frequencies stay **constant** from generation to generation: **$p + q = 1$** and **$p^2 + 2pq + q^2 = 1$**. If **$p = 0.7$** then **$q = 0.3$** ($q^2 = 0.09$, $2pq = 0.42$) → **A** P24 Q1a(xxv).
- It is the **baseline**: if frequencies **change**, evolution is happening and the forces responsible can be found P25 Q8f. "Only the strongest lion mates" = non-random mating/selection → **does not hold** the law → **FALSE** P24 Q1c(v).



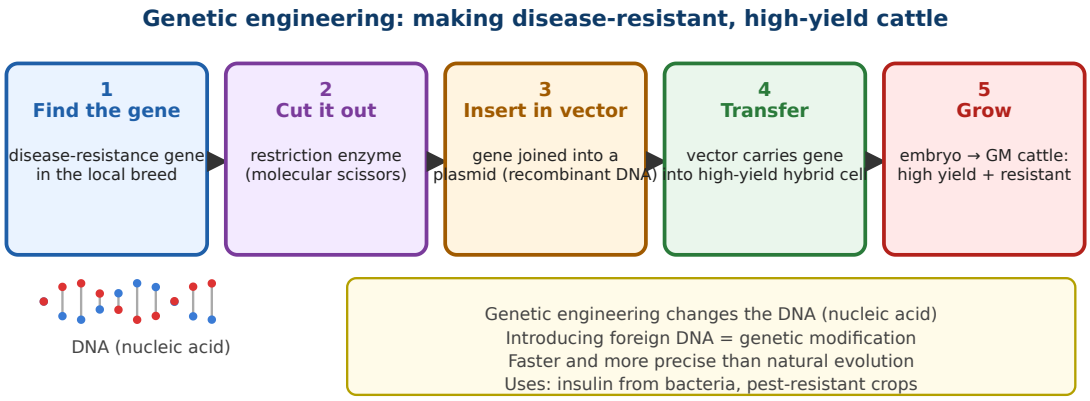
Hardy–Weinberg: genotype frequencies p^2, $2pq$, q^2 for different allele frequencies.

Image: Johnuniq, CC BY-SA 3.0, Wikimedia Commons.

Artificial selection & genetic engineering

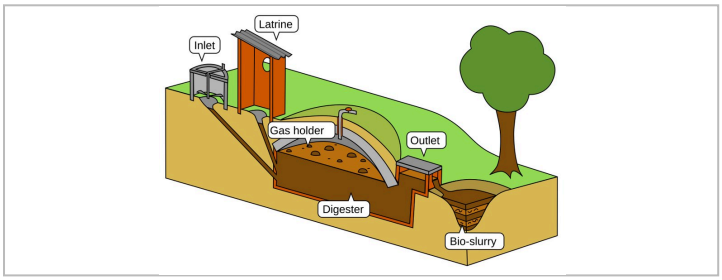
- **Artificial selection (selective breeding):** humans choose parents with wanted traits. Breeder picks **cat 3 (low shedding) and cat 1 (calm)** → **A** P24 Q1a(xxiii). **Hybrids** (e.g. hybrid corn) combine good traits.

- **Genetic engineering** (recombinant DNA) **modifies DNA (nucleic acid)** P23 Q1a(xxiv): D to **introduce desirable traits** P25 Q1a(xiv): D; introducing foreign DNA = **genetic modification** P25 Q1d(iv): b. It is **faster and more precise than natural evolution → TRUE** P24 Q1c(iv).



Steps of genetic engineering: cut the gene, insert it into a vector, transfer, grow the new organism.

Useful microbes



A biogas plant: bacteria digest waste without oxygen, giving methane gas and slurry manure.
Image: Tkarcher, CC BY-SA 4.0, Wikimedia Commons.

- **Bioremediation:** microbes clean up oil spills and waste water P23 Q1a(ii): B P25 Q1d(iii): d. **Anaerobic digestion:** bacteria break down organic waste **without oxygen** into **biogas (methane)** and slurry manure → **B** P23 Q1a(xiii). In bread-making **yeast releases CO_2 (not oxygen)** which raises the dough → **FALSE** P23 Q1c(iii).

Solved questions

P25 Q2b [2] & Q2d [2] (b) Fewer birds → **less predation** → thin-shelled snails are no longer picked off, so they survive and breed more and the **frequency of the thin-shell allele increases** (selection against it is reduced). (d) **Yes**: hybrid corn seed gives **higher yield** and **combines resistance to pests and tolerance to climate stress**; but seed is **costly and must be bought each year** and reduces crop diversity — overall a right decision.

P25 Q3d,e [3] (d) Mutations create **new alleles**; with **isolation and natural selection** they accumulate until populations **cannot interbreed**, forming a **new species**. (e) **Genetic engineering:** identify and **cut out the disease-resistance gene** of the local breed with restriction enzymes, **insert it into the DNA of the high-yield hybrid** (using a plasmid vector) and grow embryos that carry both traits, giving high-yield, disease-resistant cattle.

P25 Q5b,c [4] (b) Anaerobic digestion: (1) gives **biogas** for cooking, so food is cooked cleanly and fuelwood is saved; (2) the **slurry is a nutrient-rich fertiliser** that raises crop yield and so food supply. (c) **Situation A: ecological (habitat) isolation. Situation B: geographic isolation.**

P25 Q6b [2] & Q7d [2] (6b) **In-vitro fertilisation (IVF):** egg and sperm are **joined outside the body in a laboratory dish** and the embryo is placed in the uterus. **Significance:** helps **infertile couples** to have children (and screen genetic disease). (7d) (i) **Natural selection** (resistant mutants survive and reproduce). (ii) **No**: pests **evolve resistance**, pesticide harms useful insects and **pollutes** soil and water; use integrated pest management and biological control instead.

P25 Q8f [2] Hardy–Weinberg gives the **expected allele frequencies if nothing is changing**. (1) It is a **reference (null) model**, so a change in frequency shows **evolution is occurring**; (2) it helps identify the **forces** (mutation, migration, selection, drift, non-random mating) causing the change.

P24 Q3b [5] & Q8b [4] (3b) (i) **Genetic variation** = the differences in alleles/traits among cheetahs. (ii) Without variation all cheetahs are **alike**, so one disease or environmental change can affect all, and the population **may become extinct**. (iii) A new predator: cheetahs **cannot adapt** because no individual has a trait to escape or resist, so the **population drops**. (8b) (i) **Founder effect** (a small group colonises a new area). (ii) The small colony **drifts away genetically** and isolated from the ancestors, so over time it **cannot interbreed** and becomes a new species. (iii) **Unlikely:** humans are **one species** with constant **gene flow** between populations; no barrier stops interbreeding today.

P24 Q6d [4] (i) **Not ethical:** ligers often suffer **health problems** (gigantism, short life), males are usually **sterile**, and they are bred only for **entertainment**. (ii) Ligers are **not found in the wild**, but creating them **wastes captive space and funds** that should save pure lions and tigers, and if released they would **dilute the gene pool** of natural populations.

P23 Q2d [3] & Q8d [2] (2d) (i) The event is the **flood** (a natural disaster) → **bottleneck** that isolates survivors. (ii) Graph: population steady, then **drops sharply at the flood**; the survivors either **recover** (with less variation) or the line **falls to extinction**. (8d) (i) Different **food**: A has a **thick, strong beak for seeds**, B a **thin pointed beak for insects**; natural selection favoured the beak that suited the food. (ii) With seeds scarce, species A **declines** (or its beak must change) (check).

P23 Q6c [3] & Q7b [3] (6c)(i) **Selective breeding:** cross tall, many-fruit plant A with short plant B; pick **F1 offspring that are short with many fruits**; breed them again for several generations until the trait is pure. (ii) It **raises the frequency of wanted alleles**, creating new varieties and a faster evolutionary change (but less diversity). (7b)(i) **Hardy–Weinberg law**. (ii) **Mutation creates new alleles**; useful ones are **favoured by natural selection** and spread, changing allele frequencies — evolution.

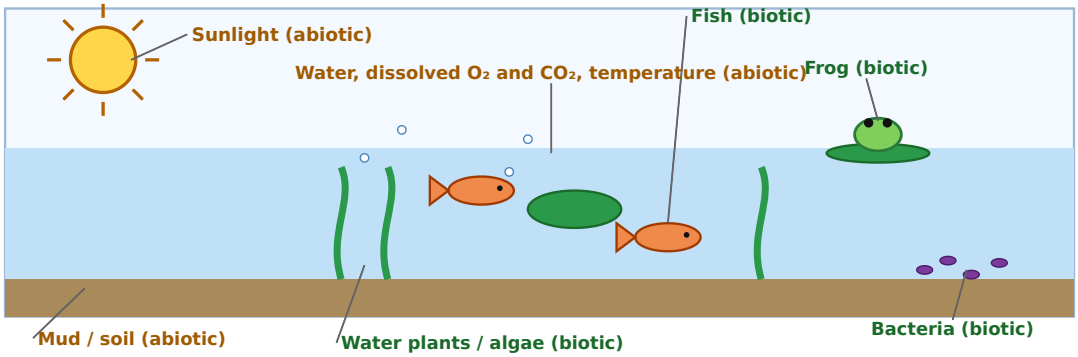
10 · ECOLOGY, BIODIVERSITY & HEALTH

P25 Q1a(x,xviii,xxii),c(iv),Q3c,f,Q4a,Q6d,e,Q8d,e,Q9e · P24 Q1a(xi–xiv),c(iii),Q5,Q6d · P23 Q1a(xiv–xvii),b(iii),Q2c,Q5b,Q6b,Q8c,e,Q9d

Ecosystem

- An **ecosystem** = **biotic** (living: plants, animals, microbes) + **abiotic** (non-living: sunlight, water, soil, air, temperature) factors **interacting** in a place P23 Q1a(xv): C. Biotic–abiotic interaction: **a tree grows in the soil** P25 Q1a(xviii): C. All the different forms of life together make **biodiversity** P23 Q1b(iii).
- **Producers (autotrophs)** are always at the **base of a food pyramid** P24 Q1a(xiv): A; then herbivores, carnivores; **decomposers** recycle nutrients.

A pond ecosystem: biotic and abiotic components



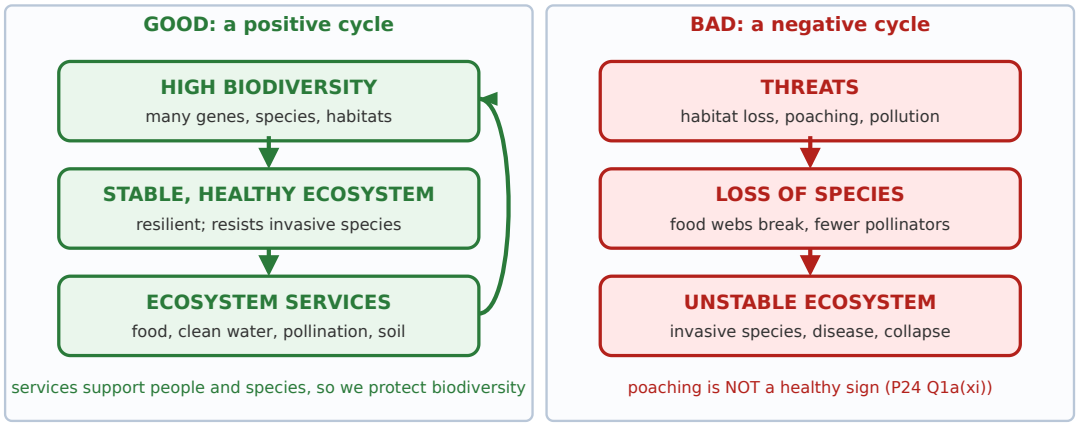
Plants need sunlight, water and CO_2; fish and frogs need dissolved oxygen and water; bacteria need mud and warmth: every biotic component depends on the abiotic ones (P23 Q9d).

A pond ecosystem: four biotic and four abiotic components that depend on each other (P23 Q9d).

Biodiversity

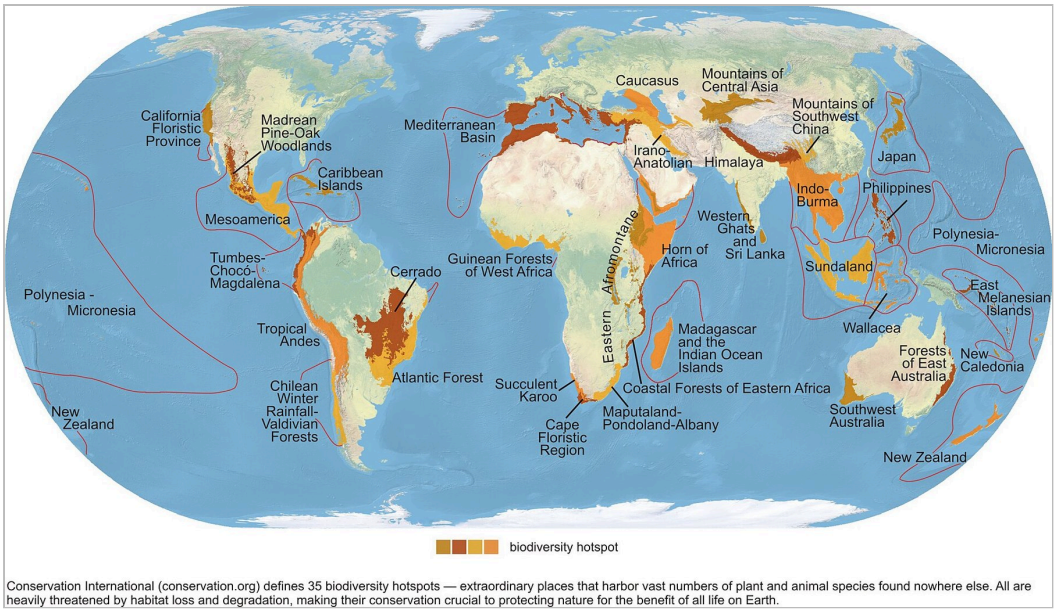
Level	Example
Genetic diversity	different breeds of dogs; DNA variation in a population
Species diversity	various species of birds in a forest
Ecosystem diversity	**forest, desert and lakes** P25 Q1a(xxii): B

Biodiversity and the health of the ecosystem



Biodiversity and ecosystem health reinforce each other (draw this for P24 Q5c).

- **A healthy ecosystem needs biodiversity:** it keeps **food webs stable**, **recycles nutrients**, supports **pollination**, and makes the ecosystem **resilient and resistant to invasive species and diseases** P25 Q3f. **Poaching is not** an example of this relationship → **A** P24 Q1a(xi).
- **Biodiversity hotspot** = a region with **very high numbers of species, many found nowhere else (endemic)**, that is **under serious threat**. There are **34** hotspots; the **Eastern Himalaya** (with Bhutan) is one P23 Q2c.
- **Endangered species** = a species at **very high risk of extinction** in the wild (tiger, red panda, golden langur) P25 Q8e. Bhutan supports many because of its **diverse habitats** → **TRUE** P25 Q1c(iv). The tiger graph falls steeply because of **poaching and habitat loss** → **C** P25 Q1a(x).



The Eastern Himalaya biodiversity hotspot.
Image: Conservation International, CC BY-SA 4.0, Wikimedia Commons.

Conserving biodiversity in Bhutan

- The **Constitution**: at least **60% forest cover for all time**; a network of **national parks, wildlife sanctuaries and biological corridors** (about half the country); laws against **hunting and wildlife trade**; **community forests**; **eco-tourism** with the Sustainable Development Fee; awareness programmes P25 Q3c (any two).
- **Will Bhutan remain a hotspot? Yes, if** it keeps the forest law and protected areas and controls **climate change, invasive species, poaching, hydropower and road building** P23 Q2c(ii).

Threats & invasive species

- **Threats: habitat destruction and deforestation**, pollution, poaching, **invasive species**, climate change, **high dams** (loss of biodiversity, loss of habitats) P23 Q1a(xvi,xvii) (check: the paper copy bolds option C for xvii). To protect biodiversity we should minimise **destruction of habitat**, pollution and invasive species P23 Q1a(xvi): C (check).
- **Deforestation, competency-based measure: work with local communities** on agroforestry and give standing forests an economic value → **D** P24 Q1a(xiii). **Kitchen waste** turned into manure by decomposers (figure) = **recycling** → **A** P24 Q1a(xii) (check).
- **Invasive species** = a **non-native species that spreads quickly and harms native species, the ecosystem and the economy** P25 Q6d(i) P24 Q5b(i). Bhutan examples: **Lantana camara, Ageratina adenophora (crofton weed), polka dot plant, Mikania**.



Lantana camara, a highly invasive plant in Bhutan.
Image: Alvesgaspar, CC BY-SA 3.0, Wikimedia Commons.

Health: HIV, COVID-19 and lifestyle

- **HIV** spreads by **unprotected sex, sharing needles or shaving blades, infected blood, mother to child**; **hugging does not** spread it → **A** P23 Q1a(xiv). **Reduce stigma:** awareness/education campaigns, testing with counselling, laws and workplaces that do not discriminate, support groups, involve people living with HIV P25 Q6e.
- **COVID-19** (novel coronavirus): being vaccinated **does not make mass gatherings without masks safe → FALSE** P24 Q1c(iii).
- **Beneficial bacteria** in the body: gut bacteria **help digestion, make vitamins and crowd out harmful germs** P25 Q4a.

Solved questions

P25 Q3c,f [3] (c) **Two strategies:** (1) the Constitution keeps at least **60% of the land under forest**; (2) a **network of protected areas and biological corridors** (also hunting ban, community forestry, eco-tourism). (f) A diverse ecosystem has **many linked species**, so if one is lost others fill its role and the ecosystem stays **stable and healthy**.

P25 Q6d [2] & Q6e [2] (d)(i) An **invasive species** is a **non-native organism that spreads rapidly and harms native species and the environment**. (ii) Without awareness the plant **spreads to more gewogs**, **crowds out native plants and crops**, lowers yields and costs much more to remove. (e) Stigma: (1) **run awareness programmes** that explain HIV does not spread by touch or sharing food; (2) **encourage testing and counselling** and let people with HIV share their stories.

P25 Q8d [2], Q8e [1], Q9e [2] (8d) **Not if done without care:** extraction should continue **only with sustainable limits** (environmental impact assessment, restoration), because **biodiversity loss is irreversible**, but a country also needs development — balance both. (8e) A species **at a very high risk of becoming extinct** in the wild. (9e) Prepare for outbreaks by **strengthening health services, testing and tracing**, keeping **stocks of masks, medicines and vaccines**, regular **awareness and hygiene education** and quick **quarantine** rules.

P24 Q5 [8] (a)(i) **Two challenges of mineral management: over-extraction** because minerals are **non-renewable**, and **damage to forests, rivers and soil** (pollution, landslides); also illegal mining. (ii) **Solution: sustainable mining rules** with environmental impact assessment, recycling and land restoration. (b)(i) An **invasive plant** is a non-native plant that spreads fast and harms the ecosystem. (ii) It **outcompetes native plants**, **reduces grazing land and crops**, and **lowers biodiversity**. (iii) Bhutan: **uprooting before seed set**, **awareness and community campaigns** by the National Plant Protection Centre (also surveys, quarantine rules). (c) **Draw** a two-way cycle: **High biodiversity → stable, productive ecosystem → services (food, clean water, pollination) → supports species**; and **loss of biodiversity → unstable ecosystem → more loss** (see figure).

P23 Q2c [3] & Q5b [2] & Q9d [1] (2c) (i) A **biodiversity hotspot** is a region with a **very large number of species, many endemic, but under threat**. (ii) **Yes** — 60% forest cover and protected areas conserve habitat; **but** climate change, invasive species and development must be controlled. (5b) (i) The problem is the **rapid spread of an invasive alien plant (crofton weed)**: from 5 to 17 dzongkhags. (ii) Consequences: **native plants and biodiversity are lost**; **farm and pasture yields fall** (also higher control costs). (9d) Pond ecosystem: **biotic** — algae/water plants, fish, frogs, bacteria; **abiotic** — sunlight, water, oxygen/CO_2, temperature. Plants need sunlight and CO_2; fish and frogs need dissolved oxygen and water.

P23 Q6b / Q8c / Q8e (6b) H. pylori: see Digestion. (8c) **No**, a person with HIV is not a threat: (1) **HIV does not spread by hugging, talking or sharing food**; (2) with **antiretroviral treatment** they stay healthy and can work normally. (8e) Reduce timber extraction by (1) using **alternatives** (bamboo, rammed earth, steel) and **recycling/reusing timber**; (2) **strict permits and planting trees** (afforestation).

SECTION A ANSWER KEY — 2025, 2024 & 2023

every MCQ, blank, true/false and match in one place · (check) = figure too small or answer reasoned, confirm with your teacher

2025 — Q1(a) MCQ [25]

No.	Ans	Why (short)
i	**C**	mitochondria absent in prokaryotes
ii	**D**	water from high to low solute concentration
iii	**D**	experiment has $NaHCO_3$ (CO_2), control only water
iv	**D**	phloem parenchyma stores and conducts food
v	**B**	less pancreatic enzymes: carbs, proteins, fats
vi	**C**	haemoglobin carries oxygen
vii	**C**	abscisic acid closes stomata
viii	**C**	obesity and hypertension harm kidney
ix	**C**	Helicobacter pylori
x	**C**	poaching and habitat loss
xi	**C** (check)	chromosomes with chromatids at poles: anaphase I
xii	**B**	law of dominance
xiii	**C**	black rabbits camouflaged on dark soil
xiv	**D**	introduce desirable traits and diversity
xv	**D**	gene flow = exchange of alleles
xvi	**C**	II splitting of water + IV ATP
xvii	**B**	additives cause lifestyle disease
xviii	**C**	tree (biotic) in soil (abiotic)
xix	**D**	mitosis: somatic cells, identical
xx	**D**	X dilute, Y concentrated, Z selectively permeable
xxi	**C**	receptor → sensory → inter → motor → effector
xxii	**B**	forest, desert, lakes = ecosystem diversity
xxiii	**C**	less gastric juice: protein digestion falls
xxiv	**A**	NADPH reduces CO_2 to glucose
xxv	**B**	RR × rr: 100% red

2025 — (b) blanks, (c) T/F, (d) match

	(b)	(c)	(d)
i	**translocation**	**FALSE** (limited by other factors)	**c** nucleoid
ii	**anti-A**	**TRUE**	**a** facilitated diffusion
iii	**descending**	**FALSE** (spinal cord first)	**d** bioremediation
iv	**bottleneck**	**TRUE**	**b** genetic modification
v	**malaria**	**TRUE**	**e** recessive allele

2024 — Q1(a) MCQ [25]

No.	Ans	Why (short)
i	**B**	eukaryotes have a nucleus
ii	**D**	smell of ema datsi = diffusion
iii	**A**	photosynthesis improves crop yield
iv	**B** (check)	chlorophyll content is internal
v	**B**	sieve element = oesophagus
vi	**C**	active transport of sugars in phloem
vii	**D**	chew thoroughly, eat mindfully
viii	**B** (check)	monocytes raised: infection
ix	**D**	side effects, environmental impact
x	**D**	lifestyle changes prevent kidney failure
xi	**A**	poaching is the exception
xii	**A** (check)	kitchen waste to manure = recycling
xiii	**D**	community agroforestry
xiv	**A**	autotrophs at the base
xv	**D**	46 chromosomes, 92 chromatids
xvi	**C** (check)	anaphase: chromatids at poles
xvii	**C**	body cells: maternal + paternal
xviii	**B**	more common in males
xix	**C**	50% of sons
xx	**B**	homozygous dominant parent
xxi	**D**	giraffe example of natural selection
xxii	**D**	fitness = survive and reproduce
xxiii	**A**	cats 1 (calm) and 3 (low shedding)
xxiv	**A**	advantageous alleles increase
xxv	**A**	$q = 1 - 0.7 = 0.3$

2024 — (b) blanks, (c) T/F, (d) match

	(b)	(c)	(d)
i	**hypertonic**	**FALSE** (bacteria are smaller)	**a** plasma
ii	**glucose**	**TRUE** (check)	**c** red blood cells
iii	**auxin** (check)	**FALSE** (masks still needed)	**e** proximal tubule
iv	**phloem**	**TRUE**	**f** loop of Henle
v	**heterozygous**	**FALSE** (non-random mating)	**h** collecting duct (check)

2023 — Q1(a) MCQ [25]

No.	Ans	Why (short)
i	**B**	has a nucleus: eukaryotic
ii	**B**	bioremediation
iii	**B** (check)	water moves from A (more water) to B (more solute)
iv	**D**	I and III
v	**D**	NADPH and ATP
vi	**C**	temperature is limiting in cold
vii	**B**	bile emulsifies oil
viii	**A** (check)	carbohydrate and lipid, no protein: mouth
ix	**C**	leucocytes: immunity

x	**C**	computers cannot decide on their own
xi	**D**	muscle growth is the aim, not a side effect
xii	**C**	glucose reabsorbed
xiii	**B**	organic matter without oxygen
xiv	**A**	hugging does not spread HIV
xv	**C**	ecosystem
xvi	**C** (check)	habitat destruction (B, C, D are all harmful)
xvii	**D** (check)	loss of biodiversity + habitats; paper copy bolds C
xviii	**A** (check)	more sugar made, more exported; paper copy bolds C
xix	**B**	crossing over: reciprocal exchange
xx	**D**	mitosis of nearby cells
xxi	**B**	normal father × diseased mother
xxii	**C**	more in males
xxiii	**B**	environment change drives evolution
xxiv	**D**	nucleic acid
xxv	**B**	ecological (habitat) isolation

2023 — (b) blanks, (c) T/F, (d) match

	(b)	(c)	(d)
i	**flaccid**	**FALSE** (enzymes denature at 50 °C)	**d** mitosis
ii	**photosynthesis**	**TRUE**	**c** dominant allele
iii	**biodiversity**	**FALSE** (yeast gives CO_2)	**e** meiosis
iv	**HCl (hydrochloric acid)**	**TRUE**	**a** selective breeding
v	**O**	**TRUE**	**b** evolution

LAST-DAY CHECKLIST

tick when you can write or draw it without looking

- ☐ **Draw and label:** prokaryotic cell (nucleoid!), nephron, chloroplast with light and dark reaction, reflex arc, mitosis and meiosis stages, ring experiment before/after, digestive system.
- ☐ **Transport:** diffusion / facilitated / osmosis / active (energy, gradient, protein); hypotonic, isotonic, hypertonic for animal and plant cells; potato osmoscope with control.
- ☐ **Photosynthesis:** equation; grana vs stroma; ATP + NADPH; factors and the three graphs; Elodea experiment variables; temperature optimum.
- ☐ **Plants:** xylem vs phloem and their four cell types each; translocation, source and sink; girdling results; hormones (auxin/NAA/2,4-D, gibberellin, ethylene, ABA).
- ☐ **Digestion:** enzyme, organ, end product for carbohydrate, protein, fat; bile, HCl, mucus; villi; H. pylori (cause, test, treatment, why in Bhutan); constipation; lifestyle diseases.
- ☐ **Blood:** RBC, WBC, platelets, plasma; ABO donor–recipient chart; antibodies and antigens; O universal donor, AB universal recipient; blood typing test.
- ☐ **Kidney:** filtration, selective reabsorption, secretion; ADH and aldosterone; PCT, loop of Henle (descending permeable to water), DCT, collecting duct; kidney-damaging habits.
- ☐ **Nervous:** reflex pathway; spinal reflex does not use the brain first; computer analogy (4 pairs); steroid side effects (physical and psychological).
- ☐ **Genetics:** mitosis vs meiosis table; 2n → n counts (12 → 6); crossing over; Punnett squares (Tt×tt, Tt×Tt, Ee×ee, carrier × normal); incomplete dominance vs codominance; sex-linked.
- ☐ **Evolution:** natural selection steps; peppered moth; four isolation types; bottleneck and founder effect; gene flow; Hardy–Weinberg ($p + q = 1$, $q = 0.3$); artificial selection; genetic engineering steps; IVF; anaerobic digestion and bioremediation.
- ☐ **Ecology and health:** biotic vs abiotic; three levels of biodiversity; hotspot; Bhutan conservation (60% forest, protected areas); invasive species (lantana, crofton weed, polka dot) and control; HIV spread and stigma; COVID preparation.
- ☐ **In answers:** count the marks, write that many points; label diagrams with ruled lines; define the variable (independent / dependent / controlled); always **justify** with “because...”.